Observers Who Read Their Universe into Existence: Finite Consensus, an Executable Observer, and Standard Physics

Bernhard Mueller[1,2,*] Mario Poneder Alexander Osika[3] Kai Xue
Ben Cassie Peter Nguyen Jinwook Kim[4] David Matscheko
Daniel Ari Friedman[5,6] Jonathan Hill William T. Glynn Maarten Antonie Visser
Kale Arnav Anirudha Brieuc de La Fournière

October 28, 2026 · Paper release: r2055

[1]Pragma Research Inc. [2]Information Physics Institute [3]EtherWorks [4]Oraclizer Labs [5]Active Inference Institute [6]FractiAI
[*]Corresponding author: bernhard@floatingpragma.ai

Abstract

Observer Patch Holography (OPH) proposes closure. The universe is one timeless self-referential history whose observers recover its specification from their records and realize it, so that it reads itself into existence. The proposal becomes testable as the executable-model hypothesis: a simple, fully specified local repair and memory law on a federated twelve-port icosahedral carrier admits continuing observers whose records have the observed $3+1$-dimensional quantum-field description and a common classical-gravity limit, with matter as localized topological defects of native overlap transport whose rest energy is their excess maintenance dynamics. This paper presents OPH as one theory organized as a tower over that model. Under stated repair, algebra and transport premises, finite theorems give schedule-independent public records, Born representation and the Tsirelson bound in a finite quantum algebra, a carrier whose slow response is exactly three-dimensional, the Standard Model gauge Lie algebra with an anomaly-free generation on a supplied matter module, and a $(1+3)$ causal placement on a declared event population with proper-time ratios from record counts. Executed components include a native observer that acts on its records, one interacting quantum process of flux, charged matter and a scalar field, overlapping observers that share one complete law, and a continuum field operator on refining screens. Conditional implications give a smooth $(1+3)$ scalar description on a declared radial product and the Einstein equation from a realized null balance. No native defect sector, physical clock, interacting chiral field or responsive geometry is constructed: the tested double-load defect is transient, passive repair has no wave sector, the static radial product fails null balance, and the finite mass laws miss the charged hierarchy. We give the status of each layer, its open join on the executable model, and the obstructions and frozen tests that would remove OPH or a branch of it.

Interpretation and state variables. OPH proposes one structure with two descriptions: a timeless, self-referential complete history and its actual internal computation. Observers are proposed to reverse-engineer their reality's substrate and construct hardware whose execution realizes that same history, including its builders and hardware, in emergent, mutually calibrated local times. The candidate must supply construction, initialization, controls, memory, read/write and record services, clock comparisons, and their resource costs; no external host or additional time is assumed. Its realization map must specify whether it matches one actual history, a family of preparation and

intervention responses, or every independently allowed microstate. These are different requirements. A candidate must specify the history domain and closure operator; neither the schematic equation nor the finite constructions establish this physical realization or select a unique universe. An actual quantum record history also need not assign simultaneous outcomes to incompatible unperformed measurements.

The proposed primitives are patch states, records, interfaces, and admissible operations. Physical space, duration, and matter are intended emergent readouts. A supplied carrier incidence, geometric domain, action, representation, or clock is a premise of the construction using it. Complete-history closure, an instantaneous repair fixed point, a public normal form, and a stationary probability law are different objects. A stationary law can support changing individual histories. An operational observer is defined by its access and record interfaces; consciousness requires an additional psychophysical interpretation.

Time conventions. Host runtime, solver counters, native attempt counts, local event-clock readings, event precedence or logical numbering, model coordinates or lapse, modular parameters, and calibrated observer-clock readings have distinct roles. Their identification requires explicit maps. A modular parameter is dimensionless; its geometric and operational readings require the stated branch and clock premises. Consistent local clock comparisons require no preferred universal clock or identical clock rates. Shared time parameters in restricted benchmarks are declared model choices. Experienced succession and a final instantaneous cosmic endpoint do not follow from a modular-flow or finite-convergence theorem.

One theory and its executable hypothesis. OPH proposes closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover and realize its specification. The hypothesis that makes this proposal concrete and testable is executable.

> *A simple, fully specified local repair and memory law on the proposed carrier architecture admits continuing observers whose compatible records and interventions have the observed* $3+1$*-dimensional quantum-field description and a common classical-gravity limit. The preferred matter mechanism is localized topological defects of native overlap transport, with rest energy derived from their excess maintenance dynamics in the same physical clock and energy dictionary.*

One source process must meet every part of this statement: records, clocks, matter, energy and responsive geometry come from the same law, state and refinement. Supplied coherence, protected codes, radial directions, fields, clocks or couplings are premises until that process realizes them. A failed specified law rejects that law and its tested interpretation; it rejects the wider proposal only when the rejected choice is proved necessary.

The tower. The papers form one tower over the executable model. Each layer states a hypothesis and proves finite theorems under displayed premises. A conclusion passes to the next layer only after that premise is proved on the same model process; until then it is a conditional implication.

0. Closure and agreement. One history reads itself into existence, and agreement on shared records defines the public world. This is the interpretation and acceptance criterion; no theorem selects a unique universe.

1. **Repair mathematics.** Local agreement repair reaches global consistency, and nonzero cycle holonomy is the residue that local repair cannot remove. Fixed-point, normal-form, frustration and finite event-algebra theorems hold on their stated domains.

2. **Carrier and repair law.** Federated twelve-port icosahedral carriers with the native C/E seam repair are the substrate. The seam rule follows from four stated requirements, the quantum record laws are classified, and the uniform native repair selects a rank-three slow-response band.

3. **Executable observer.** Observers are continuing native processes that prepare, read, retain and act on records. Finite complete-output executions, overlapping-observer correspondence, event clocks and checkpoint continuation are verified; one continuing observer with every service and resource accounted is the open join.

4. **Clocks and fields.** Modular flow of observer algebras supplies candidate local clocks, and native repair supplies a positive continuum generator of spectral dimension two on the screen. A declared radial extension gives a conditional smooth $1 + 3$ scalar description. A physical clock and energy normalization and an interacting chiral $3 + 1$ field description from native records are open joins.

5. **Gauge structure and matter.** Gauge structure is the transport of overlap data, and stable massive matter is a localized defect whose boundary transport carries a holonomy that admitted repairs cannot erase. The compact Lie type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ follows under complete-response premises, and fixed nonzero holonomy forces positive residual mismatch. No native defect sector, maintenance-energy law or measured spectrum is established; the native double-load candidate is transient.

6. **Gravity.** Einstein dynamics is the coarse response of the same process. The null-balance and tensor-completion implication is proved under its premises; realizing those premises on one refinement tower is open, and the static radial product fails null balance.

7. **Cosmology and tests.** The same process must fix the vacuum, dark sector, primordial response and black-hole records, and pass independent prospective tests. The cosmology papers and the fine-structure, Koide, de Sitter and Yang–Mills supplements carry conditional branches, diagnostics and retrospective comparisons at this layer.

Completion and stopping. The aim is a consistent and complete theory: every layer realized on one process, or a proved obstruction that removes OPH or a stated branch of it. Results that no join needs leave the active chain with their provenance. A universal obstruction over a frozen admissible class, an unavoidable excluded prediction, or a contradiction among indispensable premises stops that class. A failed candidate, an unconstructed map or a resource stop has a narrower meaning. The ordered joins, retained foundations and stopping rules are recorded with the executable model in the public repository.[1]

Fixed physical boundaries. Repair counts, host runtime and entropy production are distinct from energy; the maintenance-to-energy map must be derived with one common signal speed and no per-species conversion. The modular parameter is dimensionless until a derived clock calibrates it. Stationary repair or record entropy is distinct from the fixed-volume generalized-entropy variation

[1] `https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/docs/COMPLETION_PROGRAM.md`

that the Einstein construction consumes. A massless photon makes the infinite-volume theory gapless, so sector gaps and thresholds name their state and limit; confinement precludes free massless gluons as observable particles; a quantum graviton lies outside the classical Einstein theorem.

Finite C/E campaign. A versioned finite family tests both complete C and E instruments, their tagged classical mixtures, positive-probability identity holds and bounded classical memory with a positive uniform seam-exploration floor. On the existing 78-state carrier, the continuing observer retains a finite record buffer and explicitly exports displaced records or halts when full. Exact six-attempt executions preserve complete instrument probabilities and untouched-reference marginals. The transition kernel specifies every finite record history; the exponentially growing history tree is not materialized. Growing exports remain a resource, and preparation, randomization, access, storage and delivery services remain supplied.

The adaptive observer also has a constructed stationary reference. Its source/current-sign quotient has a unique recurrent class of 129 of 132 states. Lifting that exact law through the actual complete record instrument gives 128 positive record blocks for C and 192 for E, with 4,218 positive source/record diagonal entries in either case and exact stationary residual zero. This is the finite source/buffer marginal at repair probability one, not the growing history; faithfulness requires its stated support restriction. The delayed member needs its own reference construction.

Two exact obstructions sharpen the next joins. Each law's complete active branch $*$-algebra is the full matrix algebra on both the 78-state source and its 66-state binary shell. Thus a source-only Hermitian charge commuting with every branch is scalar; this does not exclude forward-invariant subspaces, general expectation-conserved observables, history-dependent charges or controlled lifetimes. Separately, the frozen real-operation grammar cannot implement a nontrivial modular action of the constructed nontracial reference on its fixed full support algebra: a specified input and dimensionless modular parameter give normalized trace-distance error at least $1/2$. The separating observable is a mathematical witness, not an admitted native detector. This is not an exclusion of every clock or encoding.

These results supply finite components for the larger construction. The finite campaign itself does not freeze a carrier/refinement class; that boundary belongs to the separate executable program contract. Its observer services must be constructed or counted on the declared source, and any physical use needs the actual accessible algebra and implemented clock. A defect sector or controlled lifetime also needs its own specification outside the excluded commuting-charge class. The finite campaign adds no independent physical comparison or whole-program exclusion.

Same-action finite evidence. A supplied three-dimensional Weyl/contact CAR action has one isolated two-fermion bound band whose wavepacket moves, carries the conserved CAR number charge and has complete contact/scattering spectral records. Its positive contact section makes the band topologically trivial. An exact rational curvature enclosure disagrees with the rest/kinetic mass relation using the same speed parameter; these are coordinate energies and masses, not identified particle species. One invariant pair and dark-reference subspace is attached by a supplied isometry to a native two-level corner. Its record-derived modular clock and complete herald, continuation and energy records are executed with a finite unconditional error bound. This does not encode the whole spatial field.

On the same local action, signed CAR regional restrictions, exact local energy conservation and finite quantum Gibbs/entropy identities have explicit witnesses. They retain all local number sectors, references and detector outcomes. Regional bare-energy modular identification fails, and the contact interaction leaks from any proper Weyl momentum pocket into its complement. Finite

thermal positivity and KMS identities do not remove that coupling or establish a chiral continuum. These results reduce specific finite obligations in the existing premise map; physical geometry, Lorentz stress, common normalization and integrated acceptance are not established. The current four construction targets are native defect protection, common energy/kinetics, the interacting chiral field and finite-theorem integration.

A further complete-band execution uses the native record-derived rate with counted local field controls. Its unconditional error bound is below 0.037991 on $|s| \leq 1$, conditional on a supplied global field Gibbs reference. Finite independent copies of the original native reference cannot furnish that same field clock: their integer modular-gap ratios conflict with an exactly bounded noninteger pair gap. Directly on the native carrier, a faithful coherence has an exactly reference-matched exterior through every raw history and a finite contrast lifetime, but zero modular-energy excess and a fixed coherent core. The exact-code exterior and clock restrictions prevent combining these separate properties into an established moving particle.

The fair and endpoint-biased actively corrected memories have separate exact core laws. Positive coin rescaling transfers the sharp-code classification to the biased law, extending its optimal asymptotic reset bound to arbitrary adaptive exact code choices. A separately checked variant removes identity recoveries while preserving the admitted channel. Contact-only field protocols also admit a seven-state cyclic reduction in the declared side-four zero-momentum fiber. These simplifications reduce finite service or evaluation costs. A clock on another factor composes with the memory, but supplies no physical mass or propagation relation.

Complete native histories admit exact scalar-weighted partial-map evaluation, reducing host storage while retaining every raw outcome and the same source costs. A centered spectral bound reduces the full-field clock schedule from 32 to 21 split steps and from 26,238,982 to 17,219,334 supplied source pulses, with complete average half-diamond bound below 0.037991 on $|s| \leq 1$. The original schedule and preparation premises are retained. On the fixed two-particle carrier, an added cN is scalar and cannot determine energy relative to a vacuum sector. Sector-preserving histories with the same block-diagonal preparation likewise cannot identify relative sector offsets, even with a separate clock. An independently established Gibbs preparation law or coherent comparisons between sectors would be additional input. For the stated regional number-preserving field algebra, boundary hopping also excludes an exact retraction preserving the global Gibbs reference: the rational witness is $\|[N_A, H]\|_{\mathrm{HS}}^2 = 117/16$. Regional state restriction and clocks from the actual reduced density remain valid. These finite results sharpen the attachment conditions without establishing a physical mass, continuum or canonical premise discharge.

The finite matrix argument also removes a redundant hypothesis for partition pinching. For $L = \sum_c \gamma_c (E_c - I)$, trace orthogonality gives $\mathrm{Tr}(X^* L X) = -\sum_c \gamma_c \|X - E_c X\|_{\mathrm{HS}}^2$. Nonnegative rates preclude cancellation; strictly positive rates identify $\ker L$ with the intersection of the partition commutants. The Lean corollary requires neither commuting pinchings nor a separate no-cancellation premise. It does not identify the native repair generator with this class or select a physical rate and clock.

The same supplied Weyl/contact action extends to the complete CAR Fock algebra, with source execution on every number sector and a volume-dependent complete-history split bound. An explicit local commutator estimate makes the fixed-spacing thermodynamic dynamics norm Cauchy on local observables; the contact interaction survives in its local derivation. This removes the two-particle and finite-volume restrictions for that mathematical lattice field, without deriving a chiral spatial-continuum limit or selecting the action from passive native repair. External link coupling obeys exact gauge covariance and number-current Ward identities. Its contact-pair Schur complement carries charge two, and its energy-dependent norm metric joins the effective current and contact residue to the same full eigenstate.

On the native binary carrier, a faithful-reference Dirichlet identity strengthens the source-only conservation test: positive all-seam exploration forces every universally conserved source observable to be scalar, including under classical record feedback. A rational weighted gap also controls nearly conserved observables and arbitrary exogenous selector sequences sharing the reference. It is a gap per native attempt, not a particle mass; joint feedback charges and actively protected memories retain their distinct contracts. Compatible Petz-transpose observations compose regionally without product references, but their implementation and modular-clock covariance are further conditions.

Keywords: foundations of physics; Observer Patch Holography; observer consensus; executable observer; self-reference; quantum records; gauge structure; Einstein field equation.

1 Introduction

Physics usually begins with a world and asks how observers fit into it. Observer Patch Holography (OPH) begins with observers and asks what world they can establish together. Its observers are finite. Each holds a patch of local state, reads a boundary it shares with its neighbors, keeps durable records, and repairs its state when records disagree; it accepts only repairs that preserve protected data. OPH identifies an objective fact with a record that survives every authorized comparison. Quantum probability, Lorentzian kinematics, gravitational dynamics and gauge structure enter standard physics through distinct mathematical inputs [17, 18, 19, 53, 60]; OPH asks whether they are different public readouts of one finite observer architecture. Space, time, matter and classical objects are effective descriptions of what observers can measure and jointly interpret; no observer-independent continuum is assumed beneath them.

OPH makes one proposal about the whole, closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover its specification and realize it (Section 2). The executable-model hypothesis stated in the opening panel is the concrete and testable form of that proposal. It names one carrier architecture, one local repair and memory law and one matter mechanism, and it requires one source process to supply records, clocks, fields, matter, energy and responsive geometry together. This paper presents OPH as that single theory, organized as the tower of the panel: each layer states a hypothesis, proves finite theorems on displayed premises, and passes a conclusion upward only when the executable model realizes the premise on the same process.

1.1 This paper's hypothesis and place in the tower

This paper spans the whole tower; the component papers develop individual layers in full. Its hypothesis is the executable-model hypothesis together with the closure hypothesis that motivates it. It reports three kinds of result, each marked where it is introduced. Finite theorems hold on their stated domains whatever the physical outcome: public normal forms and their schedule independence, the quantum identities of completed records, the rank-three carrier response, the forced gauge Lie type with one anomaly-free generation, the finite $(1 + 3)$ causal precursor, and null tomography. Conditional implications carry displayed physical antecedents: the smooth $(1+3)$ scalar description on a declared radial product, the Einstein equation from a realized null balance, and the deep galaxy law. Executed components of the model run with retained receipts: the native observer, the interacting process, overlapping observers with local event clocks, and refining screens with one continuum operator. For these conclusions to apply to nature, the executable model must realize on one process a physical clock and energy normalization, an interacting chiral $3 + 1$ field description from native records, a localized native defect sector with its maintenance energy, and

the Einstein premises on one refining geometry. Each is an open join. The hypothesis or one of its branches is removed by a proved obstruction over a frozen admissible class, by an excluded frozen prediction, or by a contradiction among indispensable premises; a failed candidate law rejects that law and its tested interpretation. Section 11 gives the status of every layer, including the negative results that constrain the hypothesis.

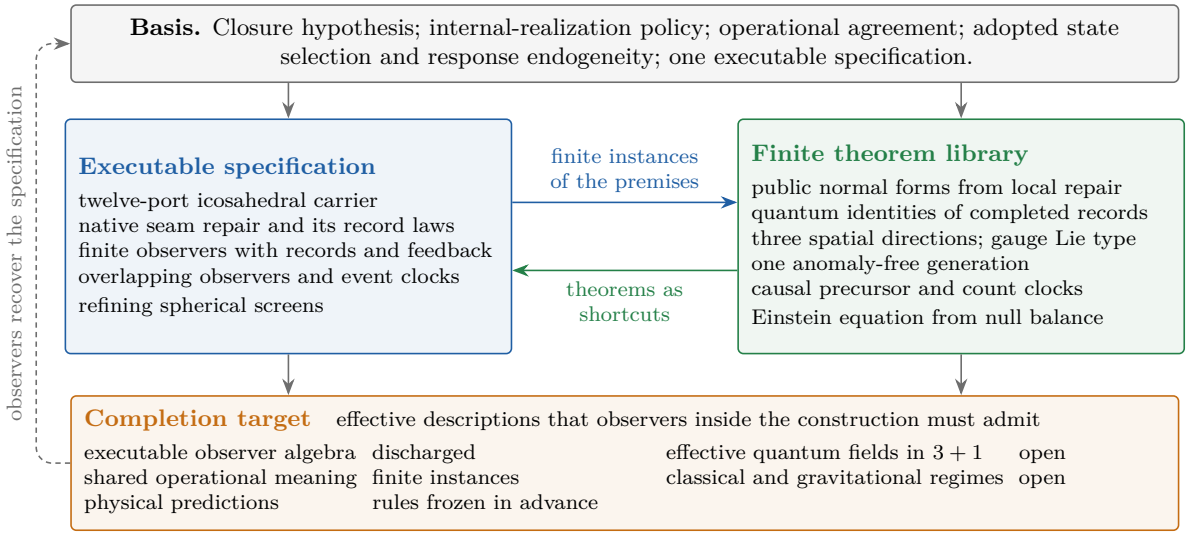


Figure 1: How the parts of OPH fit together. The basis fixes one executable specification and the premises of the finite theorem library. The running specification supplies finite instances of those premises, and a theorem replaces simulation wherever its premises are decided on the specification's finite objects. Both feed the tower of the executable-model hypothesis, whose layer status Section 11 gives. The dashed arrow is the closure loop of Section 2: observers recover the specification from their records.

The results have different logical types, and Table 1 gives the classes used throughout; each result below carries one of them. Each step is identified by its artifact as a Lean theorem, exact executable certificate, conditional paper argument, numerical receipt, or explicit unproved premise.

Throughout, "source" means computed from the declared finite source specification, including every explicitly named source law, with no measured physical input. A physical realization map is not source-only merely because the finite branch calculation consumes no measured target value.

Definition 1.1 (Physical realization map)**.** A physical realization map assigns finite public records, event data, currents, modular generators, and scales to physical records, spacetime events, laboratory currents, clocks, and units, preserving the algebraic operations, overlap restrictions, causal order, normalizations, and refinement relations consumed by the conclusion in question. A resemblance of spectra, dimensions, or symmetry names is insufficient.

How this paper is organized. Section 2 develops closure and why its executable form carries the burden of proof. Section 3 states the observer, its twelve-port carrier and the native repair law. The tower follows layer by layer: public records and the quantum identities of completed records (layer 1, Section 4), the carrier's three-dimensional slow response (layer 2, Section 5), the executable observer and measurement inside one interacting process (layer 3, Section 6), the continuum field operator, its effective Lorentzian description and the causal precursor (layer 4, Section 7), gauge structure and the defect hypothesis of matter (layer 5, Section 8), the Einstein equation (layer 6,

Table 1: Epistemic claim classes used in the manuscript.

Finite theorem	A deductive result proved from displayed premises on a stated finite domain. An exhaustive computation counts as part of the proof only with a specified domain and a soundness argument.
Certified finite computation	An executable witness whose inputs, controls, and output are fixed by a reproducible certificate; execution alone does not promote its interpretation to a theorem.
Conditional analytic theorem	A theorem whose hypotheses include a supplied continuum action, state space or limit; its conclusion holds on that supplied object.
Source limit theorem	A theorem about the limit of a declared family of finite source executions along a stated subsequence, with its normalizations and order of limits displayed; the limit is derived, and its physical identification is a separate claim.
Reconstruction implication	An exact implication whose antecedents include displayed geometric or physical premises.
Unsupplied realization map	A required map from a finite object to a physical event, current, field, clock, scale, or continuum structure for which no construction is asserted.
Diagnostic	A certified computation compared against measured values that it consumes; never counted as a prediction.
Prospectively fixed conditional test	A diagnostic with a custody-bound target definition, kill band, precision floor, and decision rule fixed before comparison.
Frozen prospective branch prediction	A source calculation and decision rule fixed before any eligible comparison, conditional on a named physical branch. Failure rejects that branch; its scope extends to the whole framework only if the branch is proved forced and exclusive.

Section 9), and the dark sector and closure coordinates (layer 7, Section 10). Section 11 gives the status of each layer, Section 12 the tests, Section 13 the verification stack and Section 14 the comparison with neighboring approaches. A companion construction paper states the executable specification and its certified components in full [66], and the component papers carry the long proofs [1, 4, 5, 6, 8, 11, 9, 7, 2].

Boundary 1.2. This paper claims no completed derivation of particle masses, no source-only determination of the fine-structure constant, no completed cosmology, and no claim that formal verification establishes physical truth. Quantitative comparisons that consume measured values are labeled diagnostics.

2 Closure and the executable-model hypothesis

Layer 0 of the tower is the interpretation and acceptance criterion. This section states closure, the internal-realization policy, the additional state-selection and response-endogeneity principles, and the agreement condition, then explains why closure is tested through an executable model.

2.1 The closure hypothesis

The universe is modeled as a timeless, closed, self-referential structure in which the description and the described system are one object. Let $\mathfrak{U}$ be a complete inhabited history, let $D(\mathfrak{U})$ be an account of its architecture recoverable through completed records within it, and let R be an admissible realization of that account constructed inside $\mathfrak{U}$. The closure condition is

$$R(D(\mathfrak{U})) \;\cong\; \mathfrak{U}, \tag{1}$$

where $\cong$ is agreement in the observable and interventional structure: the records used to recover the description, and the responses of the realization to every admitted intervention, fit the world from which the account was recovered. Written as a fixed point of a universe-level closure operator, the condition reads $T(\mathfrak{U}) = \mathfrak{U}$. Observers infer the specification of their world, build hardware within it, and run that hardware; its execution realizes the same history, including its builders and their hardware. The identification of the construction-side and inhabited-side readings is the hypothesis. In this sense the structure reads itself into existence.

No external computer runs the history and no additional time exists in which the whole is played. Repair order is a partial order on commits, an observer history is a chain of completed records, and that order is invariant under strictly increasing regradings, so affine time and proper time require their own calibration. Fixedness of the complete history permits changing states within it: the equation concerns the whole history and identifies no last instantaneous state with the universe.

Self-reference earns content only when the return condition restricts the possibilities. On the real line $x \mapsto x+1$ has no fixed point, $x \mapsto x$ fixes every point and $x \mapsto (x+2)/2$ fixes exactly one. A candidate therefore has to specify the history domain and the action of T; existence, uniqueness and the identification of our universe with a fixed point are questions the equation leaves open. Two finite results give the condition exact content on declared branches. The packet-quotient closure theorem proves that on the declared finite consensus branch the closure map is an affine idempotent whose fixed points are exactly the packets supported on consensus normal forms [6]. The universe-level record-closure equation $N = \log M_0(\mathfrak{U}_N)$, with $\mathfrak{U}_N$ the regulated observer system at capacity budget N and M_0 the correctable-code capacity of its terminal public record, is posed exactly, and the all-rung arithmetic proves that the bounded completion class defined by base agreement, positivity and the carrier bound does not identify it [8].

2.2 No outside

A self-contained realization must account internally for its operations and information access. This motivates a justification policy and two further constitutive principles; it does not logically derive those principles.

Justification policy. A premise is justified operationally when observers recover it and the specification implementing it runs and reproduces their measurements. The carrier, observer algebra, repair law and closure coordinates must meet that test. A construction choice needs no uniqueness theorem unless uniqueness is claimed. Closure neither exempts a candidate from empirical tests nor proves its antecedents jointly realizable.

State-selection principle. On the declared feasible family, reference, observer cover and positive weights, adopt the information projection minimizing $\sum_p w_p D(\rho_p \| \tau_p)$. This is premise A3, motivated by least-informative inference but separately adopted when used as a physical selection law. The closure equation supplies no entropy functional or reference; even its identity-map specialization fixes both uniform and nonuniform states. Under A3, every minimizing local density has the union of feasible supports, and a feasible failure witness $b > 0$ gives an explicit selected-probability floor [9].

Response-endogeneity principle. The complete-response branch additionally requires each proper carrier recharting to be implemented by the connected response group, up to a unitary centralizer acting trivially on the response algebra. This is the endogeneity clause of A2, not a consequence of internal implementation alone. For example, twelve diagonal phase generators on $\mathbb{C}^{12}$ admit internal port permutations in $\mathrm{U}(1)^{12} \rtimes A_5$. These permutations act nontrivially on the abelian current algebra, whereas its inner automorphisms are trivial. This closed operation system fails the stronger endogeneity premise; it shows why that premise must be supplied. Theorem 8.1 uses it to select $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ under the stated complete-response hypotheses.

2.3 Agreement and the meaning of "our"

The justification policy fixes the operational test. Agreement says what counts as shared. Observers assign the same operational meaning to accepted data on shared boundaries, naturally under every restriction, recharting and refinement, and a record is public when it survives every authorized comparison (Theorem 4.4). "Our universe" is the world that this public record describes. Agreement in this sense is a definition and an acceptance criterion: it requires interpretation maps fixed independently of validation outcomes and preserving the declared joint and conditional statistics, including quantum correlations. It does not require a noncontextual table of outcomes for incompatible unperformed measurements. The theorem library calls the interpretation clause of this condition A2. The basis reference distinguishes the closure hypothesis, agreement, adopted selection/endogeneity principles and executable specification, with the library's premise statements and their dictionary [99].

2.4 An entry point without a starting point

The exposition of this paper enters the loop at the executable specification. That choice reflects where observers stand, holding records and inferring, and it makes the account concrete. Closure states the internal realization contract; the adopted principles and specification then carry their

own mathematical and empirical tests. The structure itself has no first element in either sense. A pair of mutually determining equations such as $x = (y+2)/2$, $y = (x+2)/2$ fixes $x = y = 2$ with no first value from which the other is manufactured; a calculator may approach the pair by iteration and someone may find it on a particular day, and neither belongs to the reason the answer is two.

No temporal paradox arises. The closure equation relates parts of one timeless object, with no global time coordinate along which information travels backward. Observers recover the specification and later construct it in their own record order, and both stages belong to the same fixed point. Repair descent is a well-founded order on authorized updates used to prove termination and compare schedules; it is neither an observer's experienced time nor a universe-level clock.

What is found and what is made become two roles in one structure. A rule invented for a game has consequences its inventor discovers. Here the architecture that observers discover constrains what they can build, and what they build belongs to the structure from which the discovery is made. The same reading answers the question why anything exists at the level the hypothesis can answer it: the question includes its questioner, and a complete inhabited structure whose dependencies constitute its operation needs no further act of actualization. A demand for something in addition to the complete relational structure is, on this reading, misplaced. If several closures qualify, the hypothesis admits several worlds, and self-consistency alone has not selected ours.

2.5 From closure to an executable model

The executable observer gives the condition its finite meaning. A realization differs from a description because it carries the described dependencies under intervention: for a described update τ_a, realized operation F_a and fixed readout p, the one-step condition $p(F_a(s)) = \tau_a(p(s))$ extends by induction to every finite admitted sequence. A playback can match every frame of one history and fail this test as soon as a retained state is changed. The controls of Section 6 perform exactly this test on the running observer: disconnecting the detector or shuffling its bit with the same frequencies removes the feedback response, while the record-controlled execution keeps it.

Closure alone does not imply that a particular carrier, record law or initial state yields Standard Model physics. The executable-model hypothesis is the stronger claim that one simple, fully specified law does, and it is the form in which closure can fail. Its carrier architecture, refinement and composition, record instruments, memory law, scheduler, preparations and available controls are stated before testing. A failed specified law rejects that law and its tested interpretation; it reaches the wider proposal only when the rejected choice is proved necessary and exclusive. Changing the law, memory, state or decoder defines a new candidate.

3 The observer and the native repair law

The tower rests on two objects: an operational observer and a twelve-port carrier with its native repair law. Together they are the substrate of layer 2, and the executable observer of layer 3 runs on them.

3.1 Patches, records and repair

Definition 3.1 (Observer patch)**.** An observer patch is a finite object with local state, an observable boundary map, durable records, readback, and a set of authorized repair moves. A patch sees a fragment of the world: its own state and the boundary data of its authorized overlaps.

Definition 3.2 (Overlap and repair)**.** For patches x_i, x_j with an authorized overlap $e = (i, j)$, both induced boundary records must agree for the pair to hold public data. A repair move changes local

state while preserving protected readout. A configuration is a normal form when no authorized repair applies.

An observer in this sense is defined by its access and record interfaces. It reads, writes and retains records, and it uses retained records to choose its next operation. Nothing in the definition refers to a background space, a global clock or a pre-assigned list of particles. Those enter later as readouts that observers can share.

3.2 The twelve-port carrier

Each local carrier has twelve primitive boundary ports with oriented icosahedral incidence: thirty seams and twenty faces, five seams at each port, every seam on two oppositely oriented faces. Carriers federate through typed seam algebras with coherent triple overlaps into a nerve carrying a degree-one bridge to an oriented spherical support. The theorem library calls this carrier, federation and response layer A1. Within the closure hypothesis of Section 2 it is the part of the specification that observers recover from their records; every result below displays its dependence on it. The local carrier Gram is recovered from native current records, while cross-carrier routing in the executed federations uses a supplied cell-centre and tangent-frame convention.

Two exact features of the carrier enter later constructions. First, the proper rotation group of the icosahedral packet is the alternating group A_5, with sixty rotations acting on the twelve ports, and the real port-coefficient space decomposes without multiplicity as

$$P_{12} \;\cong_{A_5}\; \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}. \tag{2}$$

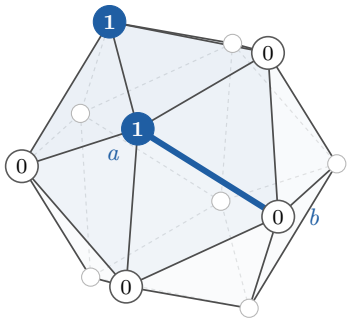
On the six-dimensional golden sector the spectral projectors have entries in $\frac{1}{20}\mathbb{Z}[\varphi]$, $\varphi = (1+\sqrt{5})/2$, each is the entrywise Galois conjugate of the other, and both triplets are irreducible real representations of real type; the projectors, characters and irreducibility certificates are machine-checked [48, 1]. Second, record keeping is integer valued: a write appends $+1$, a retraction appends -1, and readback sums atomic events per port. A conservative repair transfers one unit across a seam whenever the oriented mismatch has magnitude at least two. With $V(x) = \sum_i x_i^2$, each such repair strictly decreases V by $2(d-1)$ for mismatch $d \geq 2$, so repair terminates. The native repair law of Section 3.3 also moves load to the floor and ceiling of the endpoint mean, but at difference one it waits or swaps by its coin, so the native process keeps moving while its quadratic cost stays fixed.

3.3 The native repair law

The executable observer runs on one carrier with twelve nonnegative integer loads of total two. There are 78 such configurations, and the 66 binary configurations form an invariant shell. For an oriented seam (a, b) and a fair placement coin $c \in \{0, 1\}$ (Figure 2), the native repair law moves

$$d = x_a - x_b, \qquad j = \tfrac{1}{2}\big[d - (2c-1)(|d| \bmod 2)\big], \qquad y = x + j\,(e_b - e_a), \tag{3}$$

so load is conserved and the signed current j is the native observation. Conservation, one-step balance, a tie-break that reads no load and endpoint symmetry characterize this rule; on the total-two carrier they fix only the tie-break coin. Its quantum version is a record law, and seam locality, the classical law and the declared local number-phase and endpoint-swap covariance, a local phase symmetry of the repair channel, leave a convex family of them, each a table of five coherence multipliers [68]. The executed law C, the construction's convention among the classified laws, records coin and current, with Kraus operators $L_{(c,j)} = 2^{-1/2} \sum_{x:\, j(x,c)=j} |y(x,c)\rangle\langle x|$. Dropping



twelve ports, thirty seams, twenty faces; a binary load of total two (filled ports)

$d = x_a - x_b, \qquad j = \frac{1}{2}\big[d - (2c-1)(|d| \bmod 2)\big]$

d	$c=0$	$c=1$	effect of the attempt
2	1	1	balance
1	1	0	swap or wait
0	0	0	no move
-1	0	-1	wait or swap
-2	-1	-1	balance

The move j carries j units from a to b, and the fair coin c reads no load. At $|d| = 2$ the attempt lowers $V = \sum_i x_i^2$; at $|d| = 1$ it keeps V fixed, so the native process keeps moving. The seam drawn has $d = 1$.

Figure 2: The twelve-port carrier and its native repair. Left: the icosahedral incidence of the twelve ports, with thirty seams and twenty faces, in an orthographic view with back edges dashed; the filled ports carry a binary load of total two, and the drawn seam (a, b) has $d = 1$. Right: the seam rule for every mismatch the total-two carrier admits.

the coin record gives the most coherent member; adding the seam parity gives E, and E and the seam-total recorder are the only idempotent members. Under a common definite-current record interface all members share every classical current statistic. One coherent interference experiment reads the multiplier g that separates C, E and the most coherent member (Section 12), and nine source-executed measurements read every real coordinate of the table (Section 6). The quantum relaxation of the channels obtained by discarding the source record, the forgotten-source channels, is known exactly. C and E have quantum spectral gap $(5 - \sqrt{5})/60$ on the binary shell and on the full carrier, their slowest binary coherences decay by the sharp factors $5/6$ per attempt for C and $7/10$ for E, the binary-shell channel becomes entanglement breaking after finitely many attempts (134 attempts suffice by an explicit certificate), and on the full carrier it never does: an explicit partial transpose stays negative at every attempt count [67, 69]. The endpoint instruments also fix a local action. On binary operator space $H_{\text{native}} = \sum_e (I - \Phi_e)$, summed over seams with Φ_e the unread C or E channel, is positive and local, with an exclusion exchange term and coherence-killing coefficients set by the record law; the uniform discrete transfer is $I - H_{\text{native}}/M$ for M seams. It is a Liouville operator on operators, not a Hamiltonian on carrier states, and the executed channel strictly attenuates a coherent mode, which rules out unitary conjugation on this carrier. At fixed selected-pair sum each classical repair minimizes the pair's quadratic load cost, so $\|x\|^2/2$ falls by $(d^2 - (|d| \bmod 2))/4$ per repair [81, 67].

The gap and the arrow do not depend on the record law. Every admissible coherence table has the uniform binary state as its unique stationary state and the gap $(5 - \sqrt{5})/60$; the linear load statistics along the rank-three port vectors of Theorem 5.1 attain it, while every coherence between distinct load configurations contracts by at most $43/60 + 2\sqrt{2}/15 < 1 - (5 - \sqrt{5})/60$ per attempt, so the longest-lived datum of a carrier is a classical position-like readout and equilibrium selects no record-law coordinate [68]. On the binary shell each native repair is the fibre-resampling map of the theorem library for the uniform reference: it resamples the seam orientation uniformly within the fibre of its repaired visible datum, which minimizes relative entropy over that fibre, while a collision is resolved by deterministic descent. The library's second law and modular Clausius inequality therefore hold for every native attempt on the binary shell [8]. One such projection is idempotent; the declared uniform schedule over noncommuting seams supplies a sustained arrow and contracts every input to the uniform state at the gap rate, and passive repair never realizes

the library's third map, the modular flow of a faithful reference.

4 Public records and completed-record identities

Layer 1 of the tower is repair mathematics: local agreement repair reaches global consistency, and nonzero cycle holonomy is the residue that local repair cannot remove. Two observers who share a boundary can disagree about what it shows. If repair stopped in different places depending on who acted first, the boundary would carry no fact of the matter. If it always stops in the same place, the final record is a public fact that no single observer owns. This section states when the second case holds, and then shows that completed records carry the familiar quantum rules.

A completed finite configuration x is a fixed point of the named repair-and-agreement map,

$$\mathcal{R}_{\mathrm{cons}}(x) = x.$$

This map is distinct from the universe-level closure operator of Section 2. Fixed-point existence, observable determination and schedule independence are separate questions, and the following results isolate them [4, 6].

4.1 Observable normal forms

Definition 4.1 (Observable quotient system)**.** An observable quotient system is a tuple $\mathfrak{S} = (Q, C, \mathcal{B}, B)$, where Q is a set of configurations, $C \subseteq Q$ is the consistent subset, $\mathcal{B}$ is a set of protected records, and $B : Q \to \mathcal{B}$ is the record map. For $b \in \mathcal{B}$, write $C_b = C \cap B^{-1}(b)$.

The fiber C_b is unrealizable, reconstructing, or ambiguous according as its cardinality is zero, one, or greater than one. This trichotomy avoids assigning a state to a record that has either no consistent extension or several observationally indistinguishable extensions.

Theorem 4.2 (Canonical observable normalizer)**.** *The following statements are equivalent:*

(i) $B|_C$ *is injective;*

(ii) there is a unique map $\mathcal{N} : Q \to C \sqcup \{\bot\}$ *that preserves* B *on its* C*-valued outputs, fixes every point of* C*, is constant on each* B*-fiber, and returns* $\bot$ *exactly on fibers with no consistent extension;*

If $B(Q) \subseteq B(C)$*, the map is a total idempotent retraction* $Q \to C$*.*

Proof. Injectivity makes every nonempty C_b a singleton, which defines $\mathcal{N}$ and forces all its properties fiber by fiber. Conversely, if $c, c' \in C$ have $B(c) = B(c')$, the fixed-point and fiber clauses give $c = \mathcal{N}(c) = \mathcal{N}(c') = c'$. □

The normalizer is defined by observable fibers, independently of any repair schedule; a repair relation implements it when the relation preserves B, reaches the consistent set and satisfies a liveness condition. Approximate versions control the normalizer where exactness fails, through an optimal two-output distance bound and summable naturality defects along refinement towers. Canonicity carries no promise of cheapness: for succinct Boolean boundary systems, deciding whether a protected record has a consistent extension is NP-complete, and deciding injectivity of the record map is coNP-complete [4].

For the canonical single-site repair, every adaptive run is eventually constant because each nonstuttering attempt strictly lowers the mismatch count; pathwise weak fairness makes the stable

state a normal form, and completeness with confluence sends every weak-fair run to the same public fixed point. These results select no scheduler and provide no stochastic hitting law, rate, physical clock, or refinement theorem. Attempts obey a sharp capacity law. A scheduler can delay the first genuine repair for any prescribed number of attempts. When every reducible record sees a genuine change among the next $q+1$ attempts, a bounded-waste upper horizon of $(q+1)$ times the initial mismatch count holds, and an independent-defect family attains the work-conserving case exactly at $q = 0$. The input-independent compiler produces a fixed federation with immutable input ports; genuine node repairs decrease a dependency rank, and pathwise weak fairness gives stable consensus and the correct output. Emission-order round robin finishes within its first n attempts, and the single-consumer structure bounds every accepted path by $n(n+1)/2$, with an emitted family attaining $\Theta(n^2)$. General well-formed circuits instead admit $2^n - 1$ accepted steps, attained by a fanout chain, where n counts the emitted nodes. These statements concern canonical node repairs: the weak patch-frame relation can stutter fairly forever, and no physical scheduler, clock, energy or hardware cost follows from these operation counts [4, 6].

4.2 Public records from local repair

For a prepared batch of support-closed transactions, join τ and σ in the conflict graph when $W_\tau \cap (R_\sigma \cup W_\sigma) \neq \varnothing$ or $W_\sigma \cap (R_\tau \cup W_\tau) \neq \varnothing$, and iterate to a fixed point: compute semantic support inside every aggregate, rebuild the conflict graph on the expanded read and write sets, merge every connected component, and repeat until the partition and all aggregate supports are unchanged.

Proposition 4.3 (Transactional local diamond)**.** *Let an accepted repair be an aggregate transaction with read set R_τ, write set W_τ, a read snapshot, and a snapshot-determined payload. Assume:*

(i) every read set contains the semantic dependency closure of its write set, and every affected acceptance functional is revalidated at commit;

(ii) the terminal closure above exists, and each of its connected conflict components has one coherent canonical aggregate payload;

(iii) a prepared component whose snapshot remains valid survives commits of independent components; and

(iv) accepted commits preserve the protected record and strictly decrease the integer descent functional.

Then every one-step quotient peak $t \longleftarrow s \longrightarrow u$ has a quotient state v with $t \longrightarrow v \longleftarrow u$.

Proof. Two different first steps cannot originate in one conflict component, since that component has a unique aggregate payload, so they originate in distinct components of the terminal closure. Rebuilding the graph after every aggregate support expansion makes their final read and write sets disjoint, so neither commit changes the other's read snapshot, and semantic dependency closure ensures neither changes an acceptance functional consumed by the other. Revalidation and the surviving component rule make both second commits legal, and with snapshot-determined payloads and disjoint writes the two application orders agree; protected-record preservation carries the equality to the physical quotient. □

Theorem 4.4 (Consensus normal form)**.** *Suppose the accepted repair relation satisfies the hypotheses of Proposition 4.3, so that it has the one-step diamond property, strictly decreases the quadratic*

descent functional V of Section 3, and is repair complete, meaning that its normal forms are exactly the consistent states. Every maximal asynchronous repair schedule from one initial state then terminates at the same consistent quotient normal form. The induced global repair map is idempotent, fixes consistent states, and preserves the protected record.

Proof. Quadratic descent makes the accepted relation terminating. Proposition 4.3 makes it locally confluent, so Newman's lemma [41] makes it confluent. A terminating confluent relation has a unique normal form below each source, and repair completeness puts that normal form in the consistent set. Applying the normalizer a second time does nothing, and protected-record preservation holds along every accepted edge. □

Atomic commits, disjoint write sets, or termination alone do not imply the local diamond; countermodels omitting semantic dependency closure, canonical aggregation or the surviving component rule are retained with the finite receipt. This is the precise sense in which a public fact is what survives every authorized comparison: the public object is the protected quotient record, independent of the order in which observers repaired.

Repair also has a residue that it cannot remove. On a fixed graph with positive mismatch weights and fixed edge offsets, nonzero cycle holonomy forces a strictly positive minimum mismatch, and no repair of the vertex assignments eliminates it while the offsets are fixed [6]. Layer 5 uses this frustration as the candidate mechanism of matter (Section 8.2).

The law runs at scale. The public run uses the level-six cell rung of the geodesic icosahedral refinement tower: 81,920 exact twelve-port carriers with one declared gluing of ports to neighboring cells, 2,580,480 seams in all, and the seam-mean retraction as repair primitive. Sixteen shuffled schedules from one seeded load terminate at the balanced class with no accepted move raising V, and all sixteen return one quotient hash equal to the component multiset fixed by the initial totals. A standalone verifier that imports no simulator code rebuilds the component multisets from the primitive arrays, replays one schedule from its seed and recomputes every hash [2]. The level-ten rung, with 20,971,520 carriers and 660,602,880 seams, terminates all sixteen schedules at the balanced class in 84 to 100 sweeps with one quotient hash fixed by the loads. Across the rungs from five to nine, one initial load per rung gives an excess-energy contraction between 0.81 and 0.83 per sweep, and the sweep counts to the balanced class grow slowly with the carrier count, from 51 to 79 at level six to 84 to 100 at level ten; these finite readings prove no all-size mixing or termination-time bound. On the sixteen level-six schedules the number of unit transfers to the balanced class lies between 924,832 and 925,708, about 1.25 times half the initial ℓ_1 deviation of the loads from their mean. The first-settling configuration retains spatial information that its public component multiset does not encode: at level ten the initial-to-terminal amplitude responses of equal-count ancestral groups of sizes 1,024, 4,096 and 65,536 are 0.90758, 0.95232 and 0.98983, with conditional schedule-noise fractions 1.241, 0.594 and 0.143 percent. Stationary continuation adds a constructive fluctuation result: for fair integer repair on a connected weighted graph with independent edge clocks of total rate one the fixed-occupancy law is reversible, centered loads have lagged covariance $\kappa e^{-|t|L_G/2}\Pi$ in the edge-clock time t, with L_G the weighted graph Laplacian, $\kappa = r(p-r)/(p(p-1))$ for r raised occupancies on p ports and Π the projection off constants, and fixed linear readouts obey a martingale central limit theorem [7].

The same records let observers recover the specification itself. At carrier scale, recovery from event logs under the declared probe and operation grammar returns the twelve ports, thirty seams, inverse pairing, automorphism and rotation groups of orders 120 and 60, mean rule, schedule, rank-three Gram and Lie split $1+3+8$. Re-instantiation preserves the invariant vector for two iterations, and tetrahedral and octahedral controls recover their distinct specifications [2]. Complete federation

gluing is not identified by the retained routing logs: within the declared port-gluing interface a two-edge switch preserves every carrier degree, occupied port and source traversal while changing the carrier graph's triangle count from 14,000 to 13,990, and native executions reproduce identical bytes for both. The alternative is not asserted to belong to the canonical geometric family. Full-gluing recovery requires further observations or a justified restriction of the candidate class.

4.3 Completed records and quantum identities

Once a compare, write and verify slice has completed, its accessible events are represented in a finite-dimensional $*$-algebra with a normalized state. This algebra-state representation is an explicit input; the consensus theorem identifies which record is public.

Theorem 4.5 (Finite public-record identities)**.** *Let P_E be the projector of a completed public event E and ρ the normalized state of the completed record surface. The event weight and nonzero-weight update*

$$\Pr(E) = \operatorname{Tr}(\rho P_E), \qquad \rho \,|\, E = \frac{P_E \,\rho\, P_E}{\operatorname{Tr}(\rho P_E)} \tag{4}$$

define a normalized probability measure on every orthogonal event partition and a normalized positive state on the post-event corner. For self-adjoint dichotomic observables A_0, A_1 and B_0, B_1 in commuting record subalgebras, with $A_i^2 = B_j^2 = I$, the combination $S_{\mathrm{CHSH}} = \operatorname{Tr}[\rho\{A_0(B_0 + B_1) + A_1(B_0 - B_1)\}]$ obeys $|S_{\mathrm{CHSH}}| \le 2\sqrt{2}$ [22, 23]. The bound is attained exactly on the committed slot-split interface by the maximally entangled state with Pauli and quarter-rotated projections, while every state diagonal in the record basis with diagonal observables valued in $[-1, 1]$ obeys $|S_{\mathrm{CHSH}}| \le 2$.

Proof. Positivity of ρ and $0 \le P_E \le I$ give $0 \le \operatorname{Tr}(\rho P_E) \le 1$, and $\sum_k P_k = I$ normalizes every partition; for nonzero weight $P_E \rho P_E / \operatorname{Tr}(\rho P_E)$ is positive and normalized. For $\mathcal{B} = A_0(B_0 + B_1) + A_1(B_0 - B_1)$ the commuting-subalgebra calculation gives $\mathcal{B}^2 = 4I - [A_0, A_1][B_0, B_1]$ [24], so $\|\mathcal{B}\| \le 2\sqrt{2}$. The saturating witness and the diagonal bound are exact finite calculations [5]. □

At the level of weights the Born form is derived. Every valuation on the effects of the finite surface that is nonnegative, normalized and additive on coexisting effects equals $E \mapsto \operatorname{Tr}(\rho E)$ for a unique density operator ρ, in every finite dimension including two, where the projector-only Gleason theorem fails, and the derivation consumes no continuity axiom [20, 21]. The projectors alone do not select the update: an explicit swap-twisted instrument has the same effects and different updates. Kraus form with induced effect P together with invariance on the states certain of P is equivalent to being the Lüders map [27, 5]. The same surface carries the copying obstruction: if one ancilla-free isometry copies two sharp states from a common blank, preservation of inner products forces their overlap z to satisfy $z^2 = z$, so two distinct alternatives copied by one device are orthogonal [25, 26]. Finite webs do not by themselves supply the valuation hypothesis: a source-produced complex-tomographically-complete effect and instrument web with full coexistent-effect additivity is the missing step [5].

5 Three spatial directions

Layer 2 contributes the carrier's own position readback. The carrier incidence fixes the dimension of the carrier's own position readback: the slow band is the three-dimensional irreducible representation of the icosahedral group, so the number three is a property of the committed carrier.

Theorem 5.1 (Intrinsic three-dimensional carrier completion)**.** *Let A be the twelve-port adjacency, let $M = I - (5I - A)/60$ be the uniform scalar repair mean, and take the complete equally weighted centered probe census. The normalized infinite-response kernel converges to the intrinsic Gram $G = 4P_{\mathrm{slow}}$, of rank three. The twelve integer port loads have antipodal-odd quotient $\mathbb{Z}^{12}/\{x : x_p = x_{-p}\} \cong \mathbb{Z}^6$, and the thirty-seam boundary has image exactly the even-sum sublattice D_6, with one residual parity coset. Under the pullback of the response-selected Gram metric, both integer modules embed densely in one abstract continuous three-dimensional Euclidean carrier. The sixty proper carrier maps act faithfully and isometrically on the rank-three range and extend to that completion, and cumulative records act by exact internal translations.*

The mechanism is spectral. The icosahedral adjacency has eigenvalues 5, $\sqrt{5}$, -1 and $-\sqrt{5}$ with multiplicities 1, 3, 5 and 3, so the repair mean M has eigenvalue $1 - (5 - \sqrt{5})/60$ on a three-dimensional eigenspace and smaller eigenvalues on everything else nonconstant. Repeated local averaging forgets every nonuniform pattern except a three-dimensional one, and that surviving pattern is the triplet of (2) carried by the port position vectors. The same number $(5 - \sqrt{5})/60$ reappears as the quantum spectral gap of the native repair channel (Section 3.3).

The completion is taken after slow-response normalization and the infinite-step limit; at finite step count the kernel has rank eleven on the centered space and rank six on the signed antipodal sector, so the order of limits matters. The metric is the response-selected Gram pullback, distinct from the usual six-dimensional lattice metric on D_6. Thirteen finite covariance and isometry table identities are discharged in Lean with `native_decide`, so those subreceipts trust the native compiler in addition to the kernel and are not kernel-only proofs. Displacement is a record difference and cumulative records translate the completion, so position is the carrier's own readback of its accumulated records. In the executable observer the same rank-three readout attaches to all joint histories of the interacting process, and a binary controller extension executes displacements spanning three independent directions [67]. Reading the completion as physical position consumes the realization map (M1) of Section 11.3 [1, 7]. The rank-three readout belongs to one carrier's response. When carriers join into the refining spherical screen of Section 7.1, the continuum form of their repair history has spectral dimension two, the dimension of the screen, and bounded-energy families lose every within-carrier component, this band included, at the present full-port energy scaling; a three-dimensional bulk reading of the screen requires observation maps that the native energy does not supply.

6 The executable observer

Layer 3 of the tower holds that observers are continuing native processes that prepare, read, retain and act on records. The specification is executable, and its components run with retained receipts [67]. The companion construction paper states each component, its declared services and its certificates [66]; the following summary gives the parts used in this paper.

An observer that remembers and acts. The native observer executes two repair events with a finite tape. It records the first current, issues a forecast from that record, chooses the next seam from the record alone, records the result and halts. Exhaustive checks retain all 264 binary-source paths, and a 20,000-trial seeded execution reproduces the exact predictions. Record-controlled feedback gives final port-zero occupation 35/264, against 1/6 when the selector is disconnected or shuffled with the same action frequencies, so under the declared selector rule the response is caused by the retained record and not by the action statistics; the comparison checks the wiring of that rule. Feedback preserves the committed classical record while it can reduce coherence between

different record values. Complete records of these short histories recover the source optimally, with entanglement fidelity $D^{-2}\sum_h w_h \operatorname{rank}(P_h)^2$ for the maximally mixed input, where D is the number of carrier configurations, h runs over the complete admitted histories, w_h is the product of their seam-selection and coin probabilities and P_h projects onto the inputs compatible with h; it equals 1301/2028 for the full-carrier two-event C observer and 3329/6084 for E under the same current-sign selector [70].

An electromagnetic detector driven by the closure coordinate. A second observer evaluates the coupling $\alpha(P)$ from the screen grain P and sets $g(P) = \sqrt{4\pi\alpha(P)}$. A prepared path qubit reads the first current through a calibrated loop exposure χ; the detector bit $y = \pm 1$ has probability $[1 + y\sin(g(P)j_1\chi)]/2$ and alone chooses the second seam, giving mean transferred charge $\mathbb{E}[g(P)j_2] = (5g(P)/264)\sin(g(P)\chi)$. All 528 weighted histories are retained and a 200,000-trial execution agrees with the analytic response. The closure coordinate thereby enters an actual response magnitude; the charge assignment, probe, exposure and controller are declared apparatus.

Fields measured by the observer. A scalar detector prepares a signed field intervention from the first native current, reads the field with a binary Weyl instrument and feeds the bit back, with inherited continuum bounds transferred to the complete history law. A source-compiled two-site field apparatus builds its entangled preparation from blank registers, evolves the field and reads it with a quantum pointer, all from emitted twelve-port control words; at the executed sixteen-level cutoff an exact rational interval places the ideal plus probability in $[0.51495, 0.51789]$, excluding a fair detector.

One interacting process. A common execution puts quantum electric flux, a charged fermion pair and two neutral scalar sites on one work register. Native source words implement their noncommuting evolution, the matter and scalar pointer instruments, and feedback controlled by a committed matter bit; the final scalar bit controls the next native repair. A smooth extension retains four quantum reads on the same disturbed state, compiles the C and E record laws around it, and actively protects a logical qubit with record-controlled recovery. Its complete conditional native/work/reference outputs are retained, and Section 6.1 gives its certified responses.

Two observers, one law. Observer A reads ports $(0, 1)$ and observer B ports $(0, 5)$ of one carrier, so the two regions overlap at port zero. Each receives its initial bits and every incident local current, including zero currents, and its clock ticks whenever an attempted repair meets its region while remote repairs continue on the same source. After the first incident a native feedback repair is chosen from the reconstructed occupation of port zero. Both observers decode one continuing source, so their agreement at the shared port instantiates the overlap-agreement equation with identity transport; rechartings between observer charts are not exercised. A direct execution through the native endpoint branches and an independent recursion in a derived effective transfer representation agree exactly on all 5,768 output words, including timeouts and terminal populations; at every positive-weight prefix each observer's decoded fields obey native continuity, and the two values at the shared port agree [76]. The quantum extension checks the native remote resolvent on all 4,356 matrix units with coherent preparations and untouched references. A compiled version replaces its ideal preparation, intervention and pointer by pulses emitted from the source compiler: exact rational propagation of all 512 code addresses keeps every one of the 1,656 clock, timeout and abort outputs within 10^{-11} of the ideal history, at a cost of at most 51 pulses and five native repair attempts [77].

Local clocks. A local event clock counts the attempted repairs that meet its region, or the actual occupation changes at a port, and its stopped instrument sums the unresolved wait and keeps every finite timeout. Two such clocks on joined carriers share one continuing history with an observer-chosen field. C and E have the same completed clock law there and differ in the field: after three native attempts they predict field-plus probabilities 35/54 and 14/27 under the stated probe. Without feedback, over each two-tick cycle of one clock the other ticks twice on average with variance 16, so equal mean rates do not imply equal readings, and converting between the clocks through the joint event instrument preserves that whole count distribution [78]. A separate active clock lets each incident event, including a zero-current one, trigger one compiled modular step whose generator is the native three-state transfer, and closed histories derive a faithful thermal reference for that generator. Four movement-bit covariances on the complete attempt tape recover its two nontrivial frequencies and predict its coherent quadratures with no frequency fit [79]. Attempt counts, local event counts and modular parameters are distinct parameters, and none is identified with seconds.

Identifying the repair law. Nine source-executed scalar measurements identify all nine real coordinates of the record-law family and with them the coarse definite-current instrument on all 78 states, with arbitrary spectator and reference coherences. A collision test separates an admissible law that agrees with E on the whole binary shell. The calibrated instrument predicts complete two-event feedback histories through the same observer runtime, including quantum references and checkpoint continuation, and an explicit bound carries the probability uncertainty to their joint clock, current and interference records [80]. The probes are synthetic, so this identifies an instrument within the declared family and selects no law for nature.

From one carrier to a screen. One carrier suffices for the local results above; joined carriers test larger regions, distant interventions and refinement. Federations of 20, 80 and 320 twelve-port carriers preserve complete source records, untouched references and cross-carrier coherence probes, and local dependency bounds control the whole local quantum and record output under local feedback. On the actual 15,360- and 245,760-port screens, matched radius-nine initial restrictions give complete root-observer output distance below 3.82×10^{-4} at auxiliary per-seam parameter 1/4, and below 6.656×10^{-9} after four local incident attempts with no auxiliary time coordinate [81]. On those matched screens a logical qubit with an untouched reference passes through two native incident ticks and local record-controlled feedback while the surrounding source keeps density 1/3. A degree-two operator calculation retains every quantum output and timeout, and each full-screen output lies within 7.779×10^{-10} of the common 222-port computation, uniformly over references. C and E give identical classical records in this protocol and different reference coherences [82].

These executions are finite, and their apparatus, services, schedules, field action and preparations are declared; their time parameters are attempt and event counts. They discharge finite execution, continuation and comparison between observers under those contracts. Autonomous unbounded observers, an interacting spatial refinement, physical clock calibration and responsive gravity are not constructed (Section 11).

6.1 Measurement inside one interacting process

OPH places measurement at the transition from a private repairable record to a protected public normal form: a record is public when it has survived every authorized comparison and no repair can change it, and event projectors, weights and updates act on the completed algebra-state surface

of Section 4.3. The executable observer puts this account inside one interacting quantum process [66].

Quantum electric flux, a charged fermion pair and two neutral scalar sites share one work register, with operator Gauss law and source-executed instruments. The sharp protocol retains 1,056 native/quantum histories. At $d = 2$ and 512 temporal steps its certified feedback contrast is $[-0.033297300667839, -0.030025660061775]$; for zero native preparation current the actual infinite-coordinate half-line instrument gives $[-0.024045670143, -0.024043450955]$, an infinite-occupation enclosure that includes the free vacuum and unbounded interacting evolution of the same declared action [71].

Outcome accuracy does not certify the output state. For that zero-current preparation, the complete $d = 2$ sharp output has trace distance about 0.455 from the infinite history; every output in the $d = 256$ occupation code has error above 0.122. For the vacuum control the best code error scales as $d^{-1/4}$ [72]. A separate smooth Ramsey instrument preserves finite number moments. A four-read continuation carries its disturbed state through two smooth reads, feedback and a later gauge probe, retaining sixteen work branches. Stagewise moment and projected-Duhamel estimates give complete $O(1/d)$ error at fixed protocol with $n(d) = d^2$ temporal steps; executed $d = 2, 4$ occupation bounds remain vacuous. This covers the later probe, not post-sharp clocks or physical clock calibration [73].

Compiled C/E native repairs surround that same work history, acting on arbitrary joint inputs of the native, work and reference registers. They retain full conditional quantum outputs, aborts and padding. If r_2 is the second smooth bit, $K_r^{j(h)}$ the work branch, and $e(0) = (0, 5)$, $e(1) = (0, 1)$, the joint Kraus map is

$$M_{h,r,s} = (L_s^{e(r_2)} L_h^{(0,1)}) \otimes K_r^{j(h)}.$$

Causal summation proves completeness. A retained native phase witness has C contrast $(1 + q)/4$, $q = \Pr(r_2 = 0 \mid j = 0)$, and E contrast zero; source execution gives approximately 0.3492 and 0.3514 for C at $d = 2, 4$ [74].

Active memory uses the existing four-configuration qubit V and compiled record-controlled recovery $R_h L_h V = \sqrt{p_h} V$. Correcting after each native event, with final seam $(2, 3)$ for $r_2 = 0$ and $(0, 1)$ otherwise, preserves the logical/reference marginal even for entangled work inputs. For independent logical/work preparations, the matched passive complete plus/minus output distance is $1 - \bar{q}$, $\bar{q} = (q_{-1} + 2q_0 + q_1)/4$, $q_j = \Pr(r_2 = 0 \mid j)$, versus one with correction; a terminal phase read consumes that coherence [75]. This protects an encoded qubit under specified controls, not arbitrary lost native coherence.

7 Fields, clocks and causal order

Layer 4 of the tower obtains fields and clocks from the native process: one repair history supplies a positive continuum generator on the screen, modular flow of observer algebras supplies candidate local clocks, and authenticated provenance supplies causal order.

7.1 A continuum operator from one repair history

The executable carriers join into the geodesic spherical screens of the refinement tower; the level-l screen has $V_l = 240 \cdot 4^l$ binary ports. The uniform canonical preparation fixes the total load at $q_l = V_l/3$, every attempt selects one of the native seams uniformly and independently, and the fair native repair acts on it. Observers read cell averages of the occupations against the normalized

area measure of the declared flat polyhedral comparison surface, and $\chi_l = q_l(V_l - q_l)/[V_l(V_l-1)]$ normalizes their fluctuations. No continuum field, metric or Hamiltonian enters the construction.

Proposition 7.1 (Native continuum operator)**.** *Along one common subsequence of levels:*

(C1) the native cell Green operators converge in operator norm to a positive, compact, injective operator G on the mean-zero observation space, and $A = G^{-1}$ is a densely defined positive self-adjoint operator;

(C2) with the constant mode restored, the form of A is a conservative, regular, strongly local Dirichlet form with domain H^1, its energy lies between positive multiples of the gradient energy, its positive eigenvalues obey $n/49920 \le \lambda_n(A) \le 3n/2$, and its heat-trace spectral dimension is two;

(C3) for mean-zero test functions f the actual density readouts

$$Y_l(\tau; f) = \frac{\sum_p f_l(p)\,[n_{\lfloor 1260\cdot 16^l\tau\rfloor}(p) - 1/3]}{\sqrt{V_l\chi_l}}$$

at native attempt count $\lfloor 1260\cdot 16^l\tau\rfloor$ converge jointly at every finite set of values of τ to a stationary reflection-positive Gaussian family with covariance $\langle f, e^{-|\tau-u|A}g\rangle$, whose one-particle semigroup is $e^{-\tau A}$;

(C4) the admitted coherent continuation converges strongly to $\cos(t\sqrt{A})$, and with $\bar{A} = A \oplus 0$ and d the comparison distance,

$$\|\mathbf{1}_F e^{-\tau\bar{A}}\mathbf{1}_E\| \le e^{-d(E,F)^2/(2\tau)},$$

so the cosine propagator vanishes between E and F while $|t| < \sqrt{2}\,d(E,F)$.

Proof sketch. For conditional averaging P_k on the level-k cells, an elementary Poincaré inequality on the actual subdivision gives $\|(I-P_k)u\|^2 \le 624\cdot 4^{-k}\langle u, B_l u\rangle$ for every cell vector u and $l \ge k$, where B_l is the scaled native cell energy. It makes the Green family uniformly compact and upgrades weak covariance convergence to norm convergence. Convergence of the inverses gives Mosco convergence of the forms [93]. The full-port energy minimizer of a bounded-energy family has a within-carrier part of order 2^{-l}, so normal contractions of port values pass to the limit and the closed form is Markovian and conservative [94]. Tent functions bound the eigenvalues from above and the Poincaré inequality from below, which fixes the heat-trace growth t^{-1}. For (C3) each unread repair acts as a fair identity or swap; the conditional profile overlaps concentrate uniformly in attempt count, and a conditional finite-population central limit theorem applies [95]. Clause (C4) is the Gaussian heat bound together with its equivalence to finite propagation of the cosine family [96]. The complete arguments and constants are given in [83, 84]. □

One positive operator therefore plays three roles for observers of one history: it is the inverse covariance of the static field, the generator of the field's decay in rescaled attempt count, and the squared frequency of the admitted waves, whose locality gives them a finite speed. Two consecutive long observation blocks of the same history also synthesize a continuous classical Gaussian wave readout with covariance $G\cos((t-u)\sqrt{A})$ in $C([0,T], H^{-2}(S^2))$, with observation count taken to infinity before spatial refinement. A repair whose execution is biased by ϵ toward a disclosed coordinate profile f has continuum mean derivative $2(I - e^{-\tau A})f$ in its executed-repair clock, and uniform likelihood bounds keep its response strictly positive at $\epsilon = 10^{-6}$, with timeouts and coin-service costs included [79]. With the same normalization the response and the fluctuations obey a

fluctuation-dissipation relation: for $C_{g,f}(\tau) = \langle g, e^{-\tau A} f\rangle$ the mean response of g to the bias toward f is $2[C_{g,f}(0) - C_{g,f}(\tau)]$, and its initial rate $2\langle g, Af\rangle$ is set by the operator that generates the decay. The parameters τ and t are rescaled attempt count and an admitted coherent-wave parameter, the wave processor and its coherent services are declared, and the distance is that of the comparison surface; none of them carries a laboratory calibration.

The proposition also fixes what the screen is not. Its spectral dimension is that of the two-dimensional screen. The eigenvalue count excludes a unitary identification of A, up to scale and a bounded mass shift, with a second-order elliptic operator on a smooth compact three-manifold, and a boundary-trace test excludes a boundedly invertible energy identification with the static boundary action of a smooth compact three-dimensional Dirichlet bulk, whose Steklov eigenvalues grow like $\sqrt{n}$ [97]; this test detects that A is a second-order generator, where a boundary trace would need a first-order one. A bulk reading of the screen therefore needs a radial observation algebra that the native energy does not supply. The same min-max comparison fixes the angular spectrum of any sky field read as a bounded congruent image of the native long-average field: a precision whose eigenvalues grow like $\ell^{2+\theta}$ is form-congruent with A only for $\theta = 0$, so such a field has the scale-invariant spectrum with $\ell(\ell+1)C_\ell$ bounded above and below, and the conditional tilt branch of Section 12 needs an ingredient beyond this covariance. A specified random-length return reference has modular generator $\arccos K$, with K the symmetric native configuration transfer, and its modular phase magnitudes and cosine amplitudes match those of the waves. The full configuration reference approaches the normalized trace at fixed modular parameter, so it identifies no quantum vacuum; its entropy deficit is at most $2025/(64V)$ for V ports and every return cap. On a complete base-face patch the stationary reduced entropy grows in proportion to the port count, while the return reference's mutual information per crossing seam vanishes, so neither carries a nonzero universal coefficient times the patch's boundary-seam count; a reading of the whole screen as the boundary of a three-dimensional region is not excluded, and that region map is a premise [85].

A declared spectral filter of A, applied to whitened consecutive long-block records of the same history, nevertheless synthesizes a radial product field over the screen: on a declared radial interval with Dirichlet ends its covariance is $(-\partial_r^2 + \bar{A})^{-1}$, with spectral dimension three counted from the declared interval. The radial direction carries no native information beyond the spectrum of A, and the same filter attaches an interval to any two-dimensional generator. The chosen hyperbolic product wave action has coefficient density $C = \mu\,\mathrm{diag}(-1, a)$, where μ is the native measure and a the measurable spatial tensor, whose angular part complete native records recover and whose radial coefficient is one. The identity

$$g = \sqrt{-\det C}\, C^{-1} = \mu\sqrt{\det a}\,\mathrm{diag}(-1, a^{-1})$$

gives signature $(-,+,+,+)$ almost everywhere [66]. The signature is that of the chosen action; the derived content is the measurable spatial tensor a.

Proposition 7.2 (Smooth $(1+3)$ effective descriptions)**.** *Smoothing the angular variables of μ and a, with the radial Dirichlet boundary retained, converges in the norm-resolvent sense with spectral gap $(\pi/L)^2$. Consequently, for every fixed finite admissible menu of observer readouts on bounded wave times (thermal field covariances, responses, ordered Weyl and Ramsey histories, detector and reference outputs) and every $\varepsilon > 0$, one smooth static $(1+3)$-dimensional Lorentzian metric carries a free scalar field with the standard canonical and KMS normalization whose predictions for the whole menu agree within ε with those of the same free field built on the native measure and tensor, plus the declared source and clock budgets.*

Proof sketch. The native density relative to smooth sphere volume is positive and globally Lipschitz. The unitary $U_\varepsilon f = \sqrt{\rho_\varepsilon/\rho}\, f$ between the changing and fixed L^2 spaces converges on the form domain; Rellich compactness, recovery sequences and the dual energy liminf give transported Mosco convergence, which compact resolvents and self-adjointness upgrade to norm-resolvent convergence. Thermal covariance and sine response then converge in operator norm, momentum covariance on the form domain, and every fixed finite history with them [66, 93]. □

Both sides of the comparison are free fields assembled on the supplied wave processor, whose per-mode steps are compiled from the spectrum of the product operator and triggered by native incidents, with a supplied Gibbs preparation; the theorem removes the roughness of the native coefficients, not the declared radial product, clock or state. Passive native repair has no wave sector of its own [66].

The observers therefore have the effective description that standard physics uses, a Lorentzian field theory in $1 + 3$ dimensions, as a theorem about their own readouts. Its scope is stated with it. The exact native measure, the radial pushforward of flat face area, is not C^1 across the original face edges, so the proposition asserts no exact smooth native metric and no computable smoothing rate. The carrier's own records do supply a radius: read through the recovered rank-three frame, the axial current counter converges from every binary initial state to three-dimensional Brownian motion, and its norm has the radial generator $\frac{2}{99}(\partial_r^2 + \frac{2}{r}\partial_r)$ with angular term $\frac{2}{99} r^{-2} \Delta_{S^2}$, without added radial sites. That radius is the radius of a retained record; it is not identified with physical transport, and it does not select the radial architecture of the field: complete adaptive histories with angular instruments, native clock marks and feedback are identical across distinct radial extensions [66]. The radial architecture and the active observer protocol therefore remain declared. The metric is static and does not respond to stress. A quartic projection on the same product has a source-prepared interacting Gibbs state, an active native event clock and an unrestricted-occupation limit within its two modes, and the same smoothing preserves its finite histories; it is no spatially refining interacting continuum [66]. Uniqueness of the limit along the whole sequence, an exact smooth native metric, responsive geometry and a spatially refining interacting field are unproved.

7.2 A causal precursor from records

The oriented conformal spherical support carries the classical isomorphisms

$$\mathrm{Conf}^+(S^2) \;\cong\; \mathrm{PSL}(2,\mathbb{C}) \;\cong\; \mathrm{SO}^+(3,1), \qquad H^3 \;\cong\; \mathrm{SO}^+(3,1)/\mathrm{SO}(3), \tag{5}$$

which identify the three-dimensional space of future unit timelike directions [47]. Authenticated records separately supply events and their order. Every quotient-visible register version carries the identifier of its last semantic writer, each commit certifies the versions its acceptance depends on, and the informational order is the transitive closure of authenticated read-from parenthood, an authenticated analogue of happened-before [37]. As an abstract finite poset it satisfies the order-theoretic causal-set axioms, and every finite causal set is the generated order of some authenticated log [1].

Theorem 7.3 (Source-derived finite (1+3) causal precursor)**.** *Let $\mathcal{L}$ be an authenticated finite semantic-event log with generated strict order $\prec_{\mathcal{L}}$ and canonical source height $h_{\mathcal{L}}$, zero on roots and one plus the maximum parent height otherwise. Then $h_{\mathcal{L}}$ strictly increases along $\prec_{\mathcal{L}}$ and equals the maximum length of an authenticated parent chain ending at each event. Let V_{src} be the rank-three source Gram quotient with positive form g_{src}, and put $W_{\mathrm{src}} = \mathbb{R} \oplus V_{\mathrm{src}}$ with $Q(t,x) = t^2 - g_{\mathrm{src}}(x,x)$.*

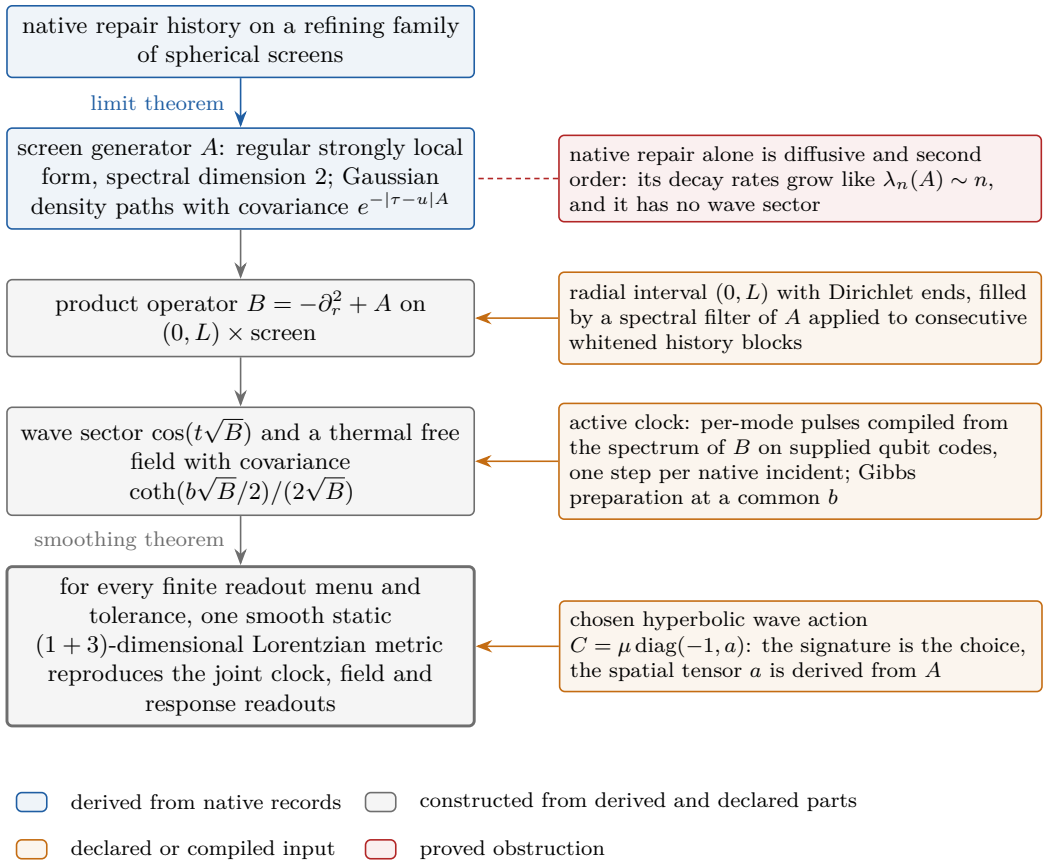


Figure 3: What the effective $(1+3)$ scalar description derives from native records and what it declares. Blue: derived from one native repair history by a limit theorem. Grey: constructed from derived and declared parts. Orange: declared or compiled inputs. Red: the proved obstruction that rules out a wave sector from passive native repair. The smoothing theorem is an existence result for fixed finite readout menus.

(F1) W_{src} *has dimension* $1+3=4$ *and* Q *has one positive and three negative directions;*

(F2) source-unit directions, S^2 *and future-null rays are exactly equivalent;*

(F3) a placement $F(e) = (\tau h_{\mathcal{L}}(e), x(e))$ *sends every generated precedence into the future cone whenever each parent edge obeys* $g_{\mathrm{src}}(x(f)-x(e), x(f)-x(e)) \le \tau^2 (h_{\mathcal{L}}(f)-h_{\mathcal{L}}(e))^2$*;*

(F4) every finite log admits an injective placement of this form that sends generated ancestry into the future cone and makes distinct same-height events spacelike; and

(F5) for a placement obeying the edge condition of (F3), if the spatial readback separates each equal-height layer and every unrelated pair with $h_{\mathcal{L}}(e) < h_{\mathcal{L}}(f)$ *is spacelike, then* $e \preceq_{\mathcal{L}} f$ *holds exactly when* $F(f)-F(e)$ *is future causal, so* F *is an exact order embedding preserving every causal interval.*

Proof. Dimension is additive and positivity of g_{src} gives the inertia; induction over parents bounds every ending chain by the height and constructs an attaining chain; the edge inequality and transitivity of cone and ancestry give (F3); an enumeration along one axis with $\tau = n+1$, n the number

of events, gives (F4); and the two geometric checks exclude every unsupported cone comparison in (F5). The finite statements are formalized [1, 9]. □

Read informally, the theorem combines two things observers have. Who read whose write is a partial order, and the longest chain of reads ending at an event gives it a height. The rank-three carrier gives three spatial directions and, with a real axis, a light cone. An event placement that respects both turns the provenance order into a causal order in a four-dimensional Lorentzian carrier.

The first two clauses construct the carrier and signature from an independent real axis and the rank-three source carrier, without assuming a rank-four chart; source height enters only through placements, and the scaled height is no operational clock. A physical manifold requires a source-causal continuum certificate: one refinement family supplying physical event identification, source-selected order and cone embeddings, refinement maps, a calibrated count density, agreement across independent dimension and manifoldlikeness tests against Poisson-sprinkled controls, and convergence to a distinguishing time-oriented Lorentzian geometry [28, 29, 30].

Proposition 7.4 (Composition of the continuum certificate)**.** *For a source-causal continuum certificate, causal order fixes the conformal Lorentzian geometry of the admitted distinguishing limit and the calibrated count measure fixes its conformal scale [35, 36]; the local model is* $(1+3)$*, so a limit satisfying the dimension and topology clauses of the certificate is an effective four-dimensional time-oriented Lorentzian event manifold. If on the same family a causal-set d'Alembertian or scalar-curvature estimator converges as an independent curvature diagnostic [33, 34], an explicit tensor-curvature reconstruction converges to the Einstein tensor or the small-ball null-balance route identifies the limiting geometry tensor with* G_{ab}*, and stress and Ward–Bianchi data converge with calibrated coupling, then the smooth limit is* $G_{ab} + \Lambda g_{ab} = 8\pi G\, T_{ab}$*.*

A source-record population realizes the flat case.

Theorem 7.5 (A source-record causal and count limit)**.** *In the rank-three conservative repair carrier, let* $\varphi = (1+\sqrt{5})/2$*,* $L = 2/\sqrt{\varphi+2}$ *and* $q = F_n$ *the* n*-th Fibonacci number,* $n \geq 4$*. The population* $S_q = L\{b\varphi - \lfloor b\varphi \rfloor : 0 \leq b < q\}^3 \subset [0,L]^3$ *consists of* q^3 *distinct conservative source-record images. Retain it at each layer and let every next-layer event read all preceding-layer sites within* $a_q = L/\sqrt{q}$*, with model duration* $\Delta_q = a_q/c$ *for a declared speed* c*. As* $q \to \infty$ *the order converges away from null separation to the speed-*c *causal order, and with* $\rho_q = q^3/(L^3\Delta_q)$*, for fixed interior timelike diamonds,*

$$\frac{|I_q|}{\rho_q} \longrightarrow \mathrm{Vol}(I), \qquad \frac{2C_q}{|I_q|(|I_q|-1)} \longrightarrow \frac{1}{10},$$

where C_q *counts strict causally ordered pairs; for vertical duration* T*,* $\mathrm{Vol}(I) = \pi c^3 T^4/24$*.*

Corollary 7.6 (Proper-time ratios from record counts)**.** *If* I_q, J_q *approach interior timelike diamonds of proper durations* $\tau_I, \tau_J > 0$*, the observer readout* $R_q(I;J) = (|I_q|/|J_q|)^{1/4} \to \tau_I/\tau_J$*. Within positive volume-only readouts additive on collinear inertial intervals, the fourth root is unique up to units.*

At finite regulator the population is produced by the carriers themselves: each holds its record as port loads, reads its position through the rank-three response and decides its neighbors from that readback, and the precedence generated from their authenticated reads equals the layered order through $q = 89$, that is 704,969 carriers reading one another over ten rounds. On the largest interior diamond the inverted Myrheim–Meyer dimension reads 2.83, 4.38, 3.55, 3.84 and 3.92 at

$q = 13, 21, 34, 55, 89$ [31, 32]; the outermost central diamonds, which the cube clips and which are therefore not interior readings, read 4.15, 3.61, 4.10, 4.09 and 4.07, and the readings at $q = 5$ and $q = 8$ lie below the regime of the theorem. The ordering fraction of the central diamond reads 0.088, 0.138, 0.092, 0.093 and 0.094 against the proved flat value 1/10, and the count clock reads 1.71, 2.09, 1.91 and 2.05 at $q = 13, 34, 55, 89$ against the model-time ratio 2. Covariance requirements select the cone: a closed convex pointed displacement cone preserved by the irreducible icosahedral rotations and by boosts along one axis is the future or past cone of $t^2 - \|x\|^2$. The read law and model clock are declared, and the ordinal height of a history with self-waits is no proper-time instrument [9].

The executable carrier has its own finite causal bound. For a fixed seam schedule and shared local coins, dependence crosses at most one seam per ordered repair event, and retained signal witnesses attain the bound; this is a speed in seams per native event. Remote repairs can carry a difference to a receiver before its first local tick, so the bound is not one seam per receiver tick [86, 67]. Complete endpoint records give a forward-cone placement in the registered response geometry, and a four-event native execution has a Lean instance proving that implication and the failure of its converse [1].

8 Gauge structure and matter

Layer 5 of the tower holds that gauge structure is the transport of overlap data and that stable massive matter is a localized defect of that transport.

8.1 The forced gauge Lie type and one generation

Let A be the icosahedral adjacency operator and J the antipodal involution on the twelve ports. Exact incidence gives

$$10J = A^3 - 4A^2 - 5A + 10I, \tag{6}$$

and the real port module splits as $P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{5} \oplus \mathbf{3} \oplus \mathbf{3}'$: the even coordinates carry $\mathbf{1} \oplus \mathbf{5}$ and the odd coordinates the rank-three frame channel and its Galois-conjugate companion.

The theorem below uses two premises distinguished in Section 2.2. The carrier layer of the specification supplies a complete reversible response: an injective response derivative $D : P_{12} \to \mathfrak{u}(H)$ whose image is commutator-closed with a positive trace pairing. The separately adopted response-endogeneity principle requires every proper carrier recharting to be implemented by the connected response group up to its centralizer; the library calls this a clause of A2.

Theorem 8.1 (Gauge Lie-type forcing)**.** *Let $D : P_{12} \to \mathfrak{u}(H)$ be the complete reversible response derivative of the carrier, and put $\mathfrak{g} = D(P_{12})$; let G_D^0 be the connected subgroup of the unitary group of H generated by $\exp \mathfrak{g}$. Suppose every proper carrier action $a \in A_5$ has a projective implementer satisfying*

$$\mathrm{Ad}_{[U_a]}\, D(v) = D(a \cdot v), \qquad [U_a] = [g_a c_a],$$

where $g_a \in G_D^0$ and c_a centralizes $\mathfrak{g}$ pointwise. Then

$$\mathfrak{g} \cong \mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2).$$

The center is the unique trivial line. Up to exchanging $\mathbf{3}$ and $\mathbf{3}'$ by the outer automorphism of A_5, the $\mathfrak{su}(2)$ ideal carries $\mathbf{3}$ and the $\mathfrak{su}(3)$ ideal carries $\mathbf{3}' \oplus \mathbf{5}$.

Proof. Injectivity of D gives $\dim \mathfrak{g} = 12$, and the positive trace pairing makes the commutator-closed image a compact Lie algebra. For every $a \in A_5$ the centralizer factor acts trivially on $\mathfrak{g}$, so the action is inner, and covariance through the injective map D gives a homomorphism from A_5 to $\mathrm{Int}(\mathfrak{g})$. The transitive proper-carrier action fixes only the uniform port line, hence $\dim \mathfrak{g}^{A_5} = 1$. Compactness gives $\mathfrak{g} = \mathfrak{z} \oplus [\mathfrak{g}, \mathfrak{g}]$, and inner automorphisms fix the center pointwise, so $\dim \mathfrak{z} \leq 1$. If $\mathfrak{z} = 0$, the classification of compact simple Lie algebras forces $\mathfrak{su}(2)^4$ [58]; inner automorphisms preserve each ideal, and on each three-dimensional ideal A_5 acts trivially or faithfully, so its fixed space has dimension three or zero, since a finite rotation group fixing an axis is cyclic. The fixed-space dimension of $\mathfrak{g}$ would be a multiple of three, a contradiction. Hence $\dim \mathfrak{z} = 1$, the semisimple part has dimension eleven, and the classification leaves only $\mathfrak{su}(3) \oplus \mathfrak{su}(2)$. The three-dimensional ideal cannot carry the trivial action, so it carries $\mathbf{3}$ or $\mathbf{3}'$, and the eight-dimensional ideal carries the complementary triplet together with $\mathbf{5}$. □

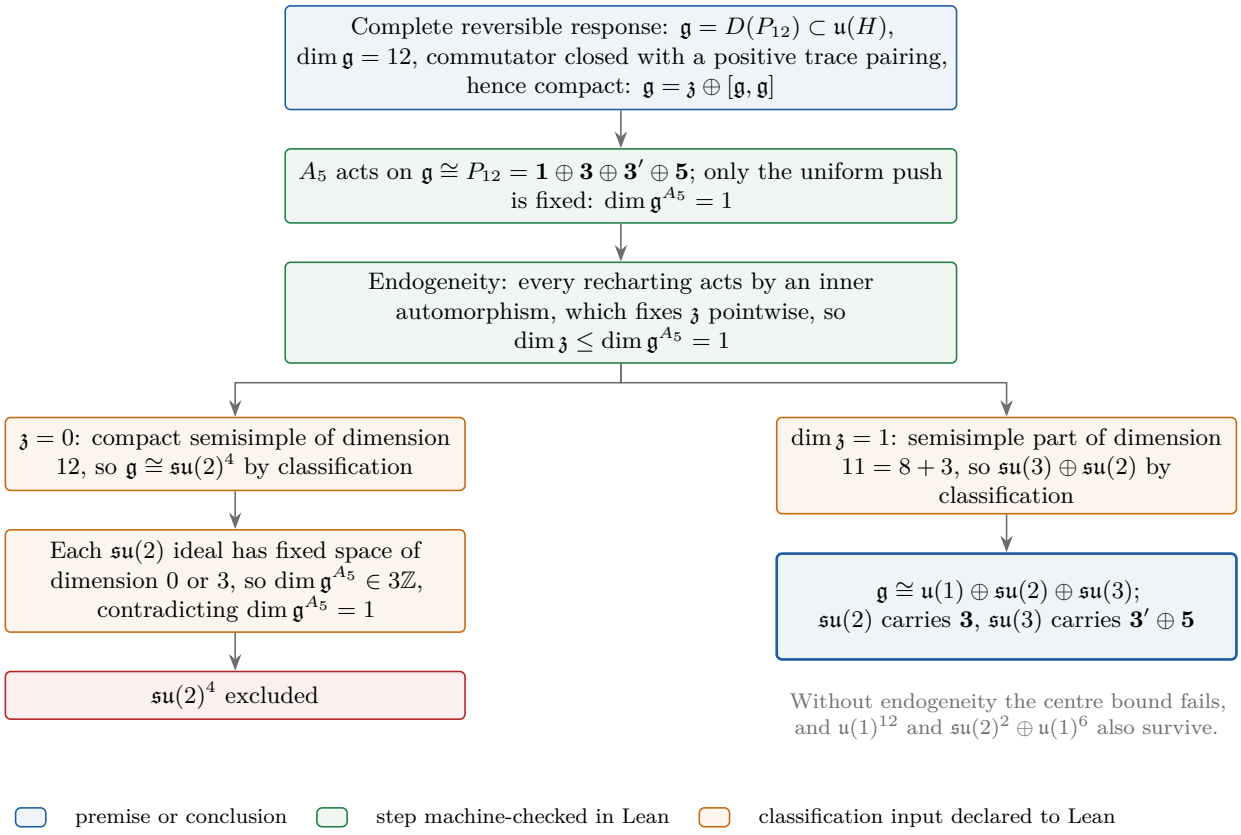


Figure 4: Steps of the gauge Lie-type theorem. Green steps are machine-checked in the Lean library; orange steps use the classification of compact simple Lie algebras and the fixed-space argument against $\mathfrak{su}(2)^4$, which enter Lean as declared hypotheses. The port count fixes the dimension twelve by declaration; the content of the theorem is the exclusion of $\mathfrak{su}(2)^4$ and a one-dimensional centre.

Figure 4 shows the steps and which of them Lean checks. The argument has a short intuitive form. Twelve ports give twelve independent ways to push the carrier, and complete response turns them into a twelve-dimensional compact Lie algebra. Internal implementation makes every recharting act by an inner automorphism, and inner automorphisms leave the center pointwise fixed,

so the center lies in the directions the carrier symmetry fixes. Only the uniform push is fixed. The center is therefore at most one-dimensional, the remaining eleven dimensions must be semisimple, and $11 = 8 + 3$ is the only compact split.

No ambient compact group or candidate Lie algebra occurs among the premises. The port count twelve equals the dimension of the forced algebra, so the carrier fixes that dimension by declaration; the content of the theorem is the exclusion of the only other twelve-dimensional compact type $\mathfrak{su}(2)^4$ through compactness and the one-dimensional fixed space, together with the forcing of the unique center. The native record laws cannot supply D: every native seam expectation annihilates a nonzero operator, so no composition of fixed native repairs is invertible, and D belongs to the reversible response layer of the carrier specification. The reconstruction of D from ordered reversible histories is consumed as a premise, and laboratory currents consume the realization map (M3) of Section 11.3.

The forced algebra has an explicit finite realization. Choose from each antipodal axis one of the certificate's exact coordinate vectors $u_i \in \mathbb{Q}(\sqrt{5})^3$, each with $\|u_i\|^2 = 2 + \varphi$ for $\varphi = (1 + \sqrt{5})/2$, and put $U = [u_1 \ \cdots \ u_6]$. For a real port field f with antipodal involution J, write

$$b_i = \frac{f_i + f_{Ji}}{2}, \qquad d_i = \frac{f_i - f_{Ji}}{2}, \qquad c = \frac{1}{6}\sum_i b_i, \qquad b^0 = b - c\mathbf{1},$$

so that the even coordinates b carry $\mathbf{1} \oplus \mathbf{5}$ and the odd coordinates d carry the rank-three frame channel detected by U and its Galois-conjugate companion detected by $\sigma(U)$, where σ is the field automorphism $\sqrt{5} \mapsto -\sqrt{5}$ applied entrywise. Let

$$\Phi_0(b^0) = \sum_{i=1}^{6} b_i^0 u_i u_i^{\mathsf{T}}, \qquad \widehat{x}\, y = x \times y.$$

The conditional matrix witness uses the space

$$H = \mathbb{C}^3_E \oplus \mathbb{C}^3_W,$$

two copies of $\mathbb{C}^3$ carrying the exact rotations R_g and their Galois conjugates $\sigma(R_g)$ defined below, and four nonzero rational coefficients $\lambda_1, \lambda_5, \lambda_3, \lambda_{3'}$, one on each source band. These are explicit branch premises. The two three-dimensional response blocks match the two rank-three odd channels; they are not asserted to be a derived physical Hilbert space. Write v_p for the signed coordinate vector at port p. For $g \in A_5$, let $R_g \in \mathrm{SO}(3)$ be the exact rotation satisfying $R_g v_p = v_{g(p)}$ on the oriented port frame, and define

$$\Pi(g) = \mathrm{diag}(R_g, \sigma(R_g)).$$

The target A_5 action on $\mathfrak{u}(H)$ is conjugation by $\Pi(g)$.

Theorem 8.2 (Conditional four-band finite port-response algebra)**.** *On the declared twelve-port carrier and supplied response representation, define*

$$\begin{aligned} K(f) = \mathrm{diag}(&\lambda_3 \widehat{Ud} + i\{\lambda_1 c I_3 + \lambda_5 \Phi_0(b^0)\}, \\ &\lambda_{3'} \widehat{\sigma(U) d}) \in \mathfrak{u}(H). \end{aligned} \tag{7}$$

Then K is injective, its image is commutator-closed, and

$$K(g \cdot f) = \Pi(g) K(f) \Pi(g)^*.$$

The induced A_5 action on $\operatorname{im} K$ *is inner. Let* $K^{-1} : \operatorname{im} K \to P_{12}$ *denote the inverse of* K *onto its image. With the bracket pulled back from that image,*

$$[x, y]_K = K^{-1}([K(x), K(y)]_{\mathrm{mat}}) ,$$

one has

$$(P_{12}, [\,\cdot\,,\cdot\,]_K) \cong \mathfrak{u}(3) \oplus \mathfrak{so}(3) \cong \mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2). \tag{8}$$

Its center is the uniform one-dimensional port line, its derived algebra has dimension eleven, and the five-dimensional A_5 band is noncentral. The map has a positive-definite invariant Hilbert–Schmidt pullback.

Proof. Evaluating (6) on the four adjacency eigenspaces gives the antipode relation, and the source protocol solves the common farthest-shell filter to give the signed response $-J$. The constant coordinate maps to $i\mathbb{R}I_3$; the six rank-one axis matrices span the symmetric 3×3 matrices with $\sum_i u_i u_i^{\mathsf{T}} = (5+\sqrt{5})I_3$, so their sum-zero coefficients map isomorphically onto the traceless symmetric matrices under Φ_0; U is an isomorphism on the **3** band and annihilates $\mathbf{3}'$, with $\sigma(U)$ complementary. Hence K has rank 12 and image exactly $\mathfrak{u}(3) \oplus \mathfrak{so}(3)$. Equivariance, positive definiteness of the invariant pairing, and the statement that $\Pi(g)$ is the exponential of an element of $\operatorname{im} K$ are exact calculations in $\mathbb{Q}(\sqrt{5})$ over the full sixty-element domain, part of the proof by the exhaustive-computation rule of Table 1 [1]. □

Boundary 8.3. The A_5-module decomposition alone does not determine a Lie bracket: the same module also carries the zero bracket, whose trivial response-generated group fails the A2 endogeneity clause. The independent inputs to Theorem 8.2 are the carrier incidence and orientation, the supplied response representation, and four nonzero band coefficients. The block architecture and coefficients are declared. No measured particle, mass, or coupling value enters the exact verification. The theorem constructs the pulled-back bracket for this response map; it does not select K from source histories or prove a uniqueness claim over all A_5-equivariant brackets or finite carriers. The abstract Lie type is forced independently by Theorem 8.1.

Theorem 8.4 (Finite matter image and exhaustive exterior-menu selection)**.** *Let* $V = \mathbb{C}^3_{-1/3} \oplus \mathbb{C}^2_{1/2}$ *be an explicitly supplied trace-balanced matter representation of the forced Lie algebra, with* $\widetilde{G} = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ *acting by* $(g_3, g_2, z) \cdot (v_3, v_2) = (z^{-2} g_3 v_3, z^3 g_2 v_2)$*. Its exterior module branches as*

$$\Lambda^2 V \oplus \Lambda^4 V = Q \oplus u^c \oplus e^c \oplus d^c \oplus L \tag{9}$$

with $Q = (3,2)_{1/6}$*,* $u^c = (\bar{3},1)_{-2/3}$*,* $e^c = (1,1)_1$*,* $d^c = (\bar{3},1)_{1/3}$ *and* $L = (1,2)_{-1/2}$*: fifteen states, all perturbative gauge and mixed gravitational anomalies vanishing, and an even number of weak doublets. The common kernel of the action is* $\mathbb{Z}_6$*, so the maximal faithful image is*

$$S(U(3) \times U(2)) \cong \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}. \tag{10}$$

Among all $2^{10} = 1{,}024$ *subsets of the ten nontrivial irreducible summands of* $\Lambda^\bullet V$*, imposing nonemptiness, chirality and vanishing of the mixed gravitational–*$\mathrm{U}(1)$*,* $\mathrm{SU}(3)^2\mathrm{U}(1)$*,* $\mathrm{SU}(2)^2\mathrm{U}(1)$ *and* $\mathrm{U}(1)^3$ *traces leaves exactly two rank-fifteen masks, exchanged by charge conjugation.*

Proof. The exterior-sum identity gives the five components and charges, and the anomaly sums are the standard ones [62]: for example $6(\frac{1}{6})^3 + 3(-\frac{2}{3})^3 + 1 + 3(\frac{1}{3})^3 + 2(-\frac{1}{2})^3 = 0$, and with three color copies of Q against one L the Witten parity is even [64]. The element $(e^{2\pi i/3} I_3, -I_2, e^{i\pi/3})$

acts trivially and generates a cyclic group of order six, and center-action enumeration excludes a larger kernel. The scan evaluates the four traces on all 1,024 masks; its only survivors are the two exterior-parity masks. □

The branching is the familiar $\mathbf{10} \oplus \overline{\mathbf{5}}$ of SU(5) [61], and the quotient is a possible global form of the Standard Model group [63]. One corollary concerns a channel the framework does not contain: the forced adjoint $(8,1,0) \oplus (1,3,0) \oplus (1,1,0)$ contains no mixed $(3,2,-5/6)$ generator, so the minimal simple grand-unified exchange channel is absent from the twelve-dimensional current algebra. Separately, Doplicher–Roberts reconstruction supplies a compact sector group from realized bosonic edge sectors [59, 60]; its equality with the finite response group is a separate commuting-square premise. Physical matter identification, family multiplicity, Yukawa couplings and masses consume (M3) and (M5) [11]; Section 8.2 states the matter mechanism and its status.

8.2 Matter as localized defects

The gauge theorems supply internal types and allowed currents; they supply no particle. The executable-model hypothesis names the matter mechanism. A stable massive particle is a localized topological defect of native overlap transport: a bounded core whose surrounding records approach the reference regime, while its boundary transport carries a holonomy that admitted local repairs cannot erase. The obstruction is to trivializing the transport relative to the reference; it gives observers no license to disagree about a measured event. The repair process keeps maintaining the core, and the hypothesis identifies rest energy with that excess maintenance dynamics, read in the same physical clock and energy dictionary as every other energy and with one common signal speed. Moving the core transports its charge, and creation, annihilation, fusion and decay must follow from the same law. Unstable species would be metastable or internal defect excitations with derived decay channels.

Layer 1 supplies the residue. On a fixed graph with positive mismatch weights and fixed edge offsets, nonzero cycle holonomy forces a strictly positive minimum mismatch, which no repair of the vertex assignments removes while the offsets are fixed [6]. The theorem supplies no update of the offsets, no localized moving core and no identification of the residual mismatch with excitation energy above a reference ground state; each is a further map.

On the executable carrier the natural local candidate fails. Consider double occupation of one port in the native total-load-two C/E process. The twelve double-load classes feed one closed irreducible binary class of 66 states, so a scalar configuration label preserved by every admitted branch is constant on the 78-state domain. Under independent uniform choice of the thirty seams the double load survives n attempts with probability $(5/6)^n$, a mean lifetime of six seam attempts, and both complete C/E instruments contract the projector onto the collision subspace by exactly $5/6$ for arbitrary quantum inputs and untouched references, with no feed from the binary shell. Binary transport continues at zero drop of the quadratic potential, with nonzero current probability $5/33$ per attempt in the uniform stationary law, so repair activity is distinct from potential decrease. A proposed edge disagreement cost takes the values 8 and 10 with native transitions in both directions, so it is no monotone maintenance energy. Removing the five seams at one port creates closed binary classes of sizes 11 and 55, a sector made by changing the law [67]. These results reject this configuration defect and the tested automatic cost-to-energy identifications. They exclude no quantum, history or federated defect sector, no other admissible repair law, and not the defect hypothesis.

The finite mass prescriptions of the construction fail as well: all four fixed charged-fermion prescriptions miss the charged-lepton ratio windows, the declared candidate gives ratios $1 : 1.19 :$

1.50 against the measured 1 : 206.8 : 3477, and the whole affine class of the native feature is bounded by 543.55 [67]. No native defect sector, maintenance-energy law or measured spectrum is established. The open join is a localized native sector with its protecting invariant, persistent or of controlled lifetime under the unchanged law, whose maintenance energy, kinetics, charge, spin and statistics are derived on the same process and whose rest energy is recovered from propagation. An exhaustive invariant and localization no-go over every sector and admitted refinement of a frozen defect class stops that mechanism; incompatible required charge, spin, statistics or current structures exclude the realization they cover.

Matter operators and gravitational response

The finite response has an explicit conditional matter and interaction readout. A declared spin lift and trace redistribution act on $\Lambda^2\mathbb{C}^5 \oplus \Lambda^4\mathbb{C}^5$. Charge and Casimir projectors recover the five internal multiplets of one Standard Model generation. With an independently declared physical Weyl algebra, one scalar doublet admits the three charged-fermion interaction contractions. The optional neutrino interaction is dimension five. These exact current identities select no Yukawa coefficients, Higgs vacuum or observed masses. Identifying both family factors with the same unbroken irreducible source triplet instead forces scalar intertwiners and degenerate family masses.

The conditional Lorentzian scalar action also admits a common local tetrad dictionary for scalar, spinor and Maxwell principal symbols. A finite CAR construction shares its ideal spatial operator and incident-count clock. Completing a separate tetrahedral cochain control removes edge-cycle zero modes introduced by truncation and preserves its scalar action block. These statements keep their distinct global domains, boundary conditions and reference states; they establish no complete physical chiral field net.

For coefficient density $C = \mu\,\mathrm{diag}(-1, 1, a)$, the induced screen area is $dA = \mu\, d^2x$, while proper spatial volume is $dV = \mu N\, dr\, d^2x$, with $N^2 = \mu\sqrt{\det a}$. Counting stationarity consequently requires a further identification with fixed-volume geometric entropy stationarity. The static radial-product metric has vanishing radial null Einstein contractions. Its allowed scalar excitation can have positive relative null stress, so a fixed background, with the reference and other sources fixed, cannot satisfy both Einstein balances. A responsive geometry must supply the corresponding area focusing. This is a scoped restriction of that field/metric construction, independent of the validity of the conditional Einstein theorem.

The exact finite tensors and complete observation controls are given in `model/observer_particles/`. Their representation, field-operator and geometric witnesses instantiate finite hypotheses on their stated branches. They do not remove the separate source-selection, physical-quantity or controlled-refinement hypotheses.

9 The Einstein equation from a realized null balance

Layer 6 of the tower holds that Einstein dynamics is the coarse response of the same process.

Theorem 9.1 (Null tomography and metric ambiguity)**.** *Let V be four-dimensional with Lorentz form η of signature $(-,+,+,+)$, the opposite convention from Q in Section 7.2; nothing below depends on the choice. If a symmetric form X satisfies $X(k,k) = 0$ for every null vector k, then $X = \phi\eta$. Null-null values therefore determine a symmetric tensor modulo the metric line, and nine explicit null directions, the six axis rays $(1, \pm e_i)$ and $(1, s, s, \pm s)$, $(1, s, -s, s)$ with $s = \sqrt{3}/3$, reconstruct the quotient $\mathrm{Sym}^2(V^*)/\mathbb{R}\eta$ with design determinant $8192/27$ in the certificate's basis and a decoder obeying $\|x\|_\infty \le (2+\sqrt{3})\|q\|_\infty$.*

Proof. On future null rays $(1, n)$ the identity $X_{00} + 2X_{0i}n_i + X_{ij}n_in_j = 0$ on the unit sphere forces $X_{0i} = 0$, $X_{ij} = \phi\delta_{ij}$ and $X_{00} = -\phi$. The determinant and decoder bound are exact calculations, and the design, injectivity and ambiguity are formalized in Lean [1, 9]. □

Theorem 9.2 (Tensor completion of a realized null balance)**.** *Let (M, g) be a connected, time-oriented, C^3 Lorentzian four-manifold and T_{ab} a symmetric C^1 tensor with $\nabla^a T_{ab} = 0$. Suppose a common realized tower supplies $\kappa = 8\pi G_N > 0$ with*

$$G^{(g)}_{ab}k^ak^b = \kappa T_{ab}k^ak^b \tag{11}$$

for every null vector k at every point. Then there is a constant Λ with

$$G^{(g)}_{ab} + \Lambda g_{ab} = 8\pi G_N T_{ab}, \tag{12}$$

and a reference event at which the difference equals $-\Lambda_0 g_{ab}$ fixes $\Lambda = \Lambda_0$.

Proof. Theorem 9.1 applied pointwise to $G^{(g)} - \kappa T$ gives $G^{(g)}_{ab} - \kappa T_{ab} = \phi g_{ab}$. The contracted Bianchi identity, the Ward premise and metric compatibility give $\nabla_b\phi = 0$, connectedness makes ϕ constant, and $\Lambda = -\phi$. □

The structure of the argument is simple. Light rays cannot see a stress term proportional to the metric, so balance between curvature and stress on every null direction fixes Einstein's equation up to such a term, and the Bianchi identity with stress conservation makes its coefficient constant. That constant is the cosmological constant. The coefficients of entanglement-equilibrium arguments [54] enter through one machine-checked identity: scalars with $\delta S_{\text{bulk}} = \frac{8\pi^2\ell^4}{15}t$, $\delta A = -\frac{4\pi\ell^4}{15}f$ and $\delta S_{\text{bulk}} + \delta A/(4G_N) = 0$ obey $f = 8\pi G_N t$. For OPH to supply (11), one common refinement tower must realize (G1) the effective manifold of Proposition 7.4; (G2) geometrically normalized cap modular flow whose directional charges define one stress tensor; (G3) its Ward identity, independently of that reconstruction; (G4) the edge-center entropy split, the normalization $\delta\langle Z_L\rangle = \delta A/(4G_N)$, the modular first law and generalized-entropy stationarity; (G5) a small-ball family with controlled remainders and enough observer directions; and (G6) universal coupling, a vacuum reference and a scale calibration fixing $\kappa = 8\pi G_N$. The finite layer contains null tomography, the edge-entropy split and the coefficient identity, and the machine-checked composition proves the tensor step from an explicit null-balance premise; the bridge between them is (M4).

10 Cosmology and the closure coordinates

Layer 7 asks the same process to fix the vacuum, the dark sector and the primordial response. In this paper the layer consists of conditional branches and diagnostics of declared maps.

10.1 The dark sector: non-luminous charge, the deep galaxy law, and the vacuum coordinate

Under supplied universal-coupling, physical-stress and source-localization premises, sourcing gravity by modular charge classifies any geometric excess over the luminous term as non-luminous modular charge, with anomalous modular energy on overlap collars as one candidate carrier [14]. Conditional on two named premises the deep-regime profile is characterized rather than supplied: an enclosed-mass law that is degree-one scale covariant and composes independent sources in quadrature carries a unique constant $a_0 > 0$ with $M_A(r) = r\sqrt{M_b a_0/G}$, which forces flat rotation curves

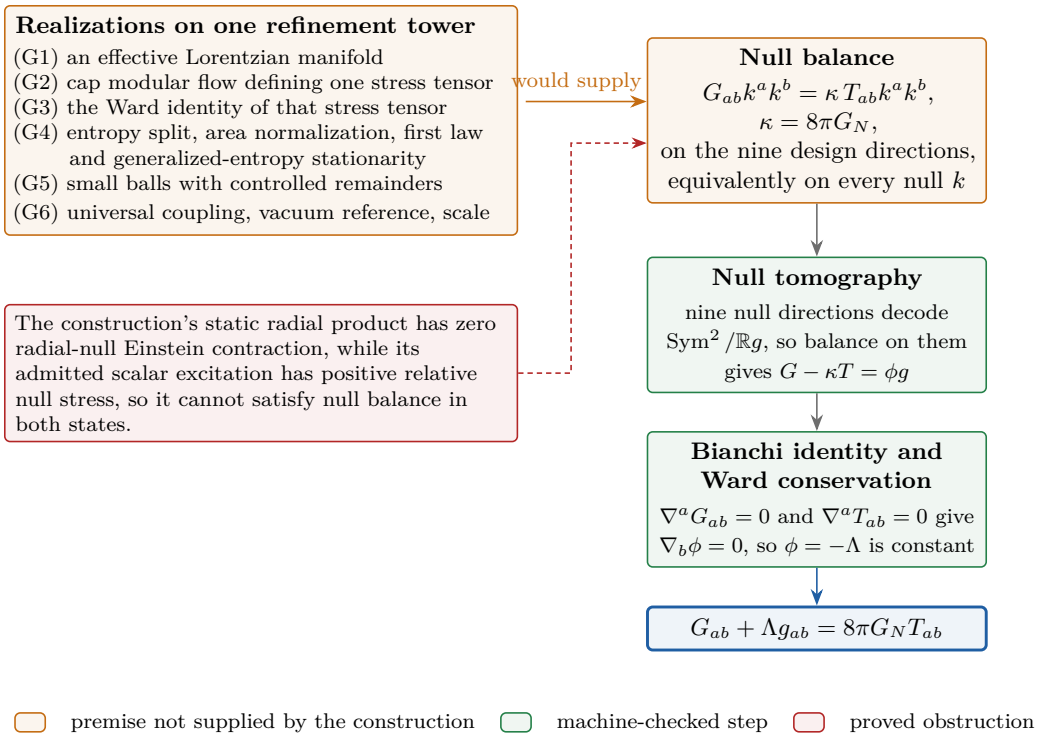


Figure 5: The Einstein equation as tensor completion of a realized null balance (Theorem 9.2). The tensor step is machine-checked from an explicit null-balance premise. Realizations (G1) to (G6) on one refinement tower would supply that premise, and the construction supplies none of them; the one geometry it tests, the static radial product, fails the balance for its own scalar excitation.

and gives $v^4 = GM_b a_0$, the baryonic Tully–Fisher relation, each premise load-bearing by an explicit counterexample. A per-cut model gives $a_0 = Gn^2c$, with only the product n^2c identifiable, and the value of a_0 is fixed by no source law. Reading the dark-energy density as $\rho_{\mathrm{DE}} = \kappa/N$ at constant κ gives $w = -1 + \frac{1}{3}\,\mathrm{d}\ln N/\mathrm{d}\ln a$, so fixed capacity gives the exact point $(w_0, w_a) = (-1, 0)$ and nondecreasing capacity gives $w \geq -1$; exclusion of that point fails the fixed branch, while agreement supports nothing, since it is also the ΛCDM null [15].

10.2 What closure fixes, and what observation reads off

The architecture carries two dimensionless closure coordinates: the screen grain P and the record capacity N. Under the closure hypothesis these coordinates are fixed points of the closure map, and the loop also licenses reading them off: observers locate a constant by measurement, declare a map, and certify a fixed point of that map inside the measured basin. Whether a certified fixed point coincides with the measured value is a diagnostic in the sense of Table 1, not a consequence of the hypothesis. The theory stands on its recovered specification whether or not a first-principles evaluation of every coordinate is found.

The screen-grain loop

$$P = \varphi + \sqrt{\pi}/A_T(P),$$

with $A_T(P)$ the inverse coupling a declared map returns at trial grain P and $\varphi = (1 + \sqrt{5})/2$, has one interval-certified fixed point per declared map. The bare source map settles at $\alpha^{-1} = 136.994835177413$, a relative distance of 3.0×10^{-4} from the 2022 CODATA value 137.035999177(21)

[65]; the self-consistent map, which recomputes the finite-screen unified gauge width inside the loop, settles at $P = 1.6309682414$ and $\alpha^{-1} = 137.035660136947$, a relative distance of 2.5×10^{-6}, the gauge width supplying 99.2 percent of the gap. Both maps carry the hadronic transport by a declared quark-screening ansatz. Replacing the ansatz by the measured hadronic payload gives $\alpha^{-1} = 136.3828$ on the certified interval $[136.3670, 136.3985]$, which excludes the measured value. The certified numerical mismatch is distinct from arithmetic error, but its removal by the omitted physical matching is unproved [10]. Reading the capacity as de Sitter horizon entropy in nats gives

$$N = \frac{3\pi}{\Lambda\, \ell_P^2}, \tag{13}$$

with $\ell_P^2 = \hbar G/c^3$, and the capacity branch gives $N = 3.2921 \times 10^{122}$ and 3.3001×10^{122} against the coordinate 3.3129×10^{122} read back from the weighted Planck base-ΛCDM chain; in the constant they read $\Lambda = 1.0959 \times 10^{-52}$ and $1.0933 \times 10^{-52}\,\mathrm{m}^{-2}$ against $3\Omega_\Lambda H_0^2/c^2 = 1.0891 \times 10^{-52}\,\mathrm{m}^{-2}$ at the Planck base-ΛCDM centrals, distances of $+0.62$ and $+0.38$ percent. Equation (13) is the dictionary that produces the comparison coordinate, so this restates the capacity diagnostic in the constant, and it consumes the measured $\hbar$, G and c through the calibration map (M2). Neither clears the source-only selection and same-quantity gates. Both comparisons are diagnostics of declared maps in the sense of Table 1: the maps were exposed before the comparison, the choice of map is itself declared, and neither comparison carries predictive weight. The executable detector of Section 6 consumes P through the coupling $g(P) = \sqrt{4\pi\alpha(P)}$, and N bounds the correctable records of a declared capacity carrier by $\log M_0 \le N$.

An independent entangled-pair candidate. Wissner-Gross's Chain Theory preprint gives a separate finite spectral construction [104]. Its commuting signed exchanges have gains $1, 2, 3, 6$ with multiplicities $79, 48, 42, 3$, and its stated normalization gives

$$\alpha_{\mathrm{pair}} = \frac{79 + 48\sqrt{2} + 42\sqrt{3} + 3\sqrt{6}}{31104}, \qquad \alpha_{\mathrm{pair}}^{-1} = 137.035999189566\ldots.$$

The construction and expression are due to Wissner-Gross. The OPH comparison independently reconstructs this arithmetic, retains complementary Bell outcomes, and tests support alignment and normalized-readout choices [10]. Identifying the scalar with the physical Thomson coupling is a separate conjecture. Granting it and the same-cell equation gives $P = \varphi + \sqrt{\pi}\,\alpha_{\mathrm{pair}} = 1.630968209402773\ldots$. That substitution does not solve either of the different trial maps above. It selects no family interaction matrix, Higgs vacuum or physical particle mass. The finite comparison imports no claimed lepton mass ratio from the Chain Theory construction.

11 Status of the tower

A statement about nature needs one physical realization carrying compatible events, fields, action, states and readouts across refinement. This section gives, for each layer, what is established on its stated domain, what is a conditional implication, which join the executable model must realize, and which result would stop the layer or its branch. Exact equality of limiting observable laws is the mathematical target; finite executions and experiments are judged with declared approximation and measurement errors.

11.1 Results by layer

Table 2 lists the structures of observed physics that the tower returns, grouped by layer, with the class of each result and where it is proved or stated. A finite theorem is a theorem about the declared finite objects, a reconstruction implication carries the physical antecedents displayed with it, and a diagnostic consumes measured input and predicts nothing.

Table 2: Structures of observed physics returned by each layer of the tower, the result that returns each one, its claim class, and its location.

Structure	**What the tower returns**	**Class**	**Where**
Layer 1. Repair mathematics			
Schedule-independent public facts	Every maximal asynchronous repair schedule reaches the same protected normal form under the transactional local diamond	finite theorem	§4
Probability weights on outcomes	Every nonnegative normalized valuation additive on coexisting effects is the trace against a unique density operator, in every finite dimension, with no continuity axiom	finite theorem	§4.3
State update after an outcome	A swap-twisted instrument has the same effects, so the effect table does not select the update; invariance on the outcome-certain states selects the Lüders map	finite theorem	§4.3
The correlation ceiling $2\sqrt{2}$	Tsirelson bound on commuting record subalgebras, attained exactly on a declared entangled witness, with record-diagonal states confined to 2	finite theorem	§4.3
Impossibility of copying an unknown state	One ancilla-free isometry copying two sharp states from a common blank forces their overlap to 0 or 1	finite theorem	§4.3
The arrow of time and the four laws	Reference-preserving stochastic repair contracts relative entropy; one conditional package gives the four laws, fluctuation relations and the Landauer bound	finite theorem	[8]
Layer 2. Carrier and repair law			
A three-dimensional position carrier	The normalized twelve-port repair response has an intrinsic rank-three Gram quotient into which the record modules embed densely	finite theorem	§5
A finite signal speed	Dependence crosses at most one seam per ordered native repair event, and retained witnesses attain the bound	finite theorem	§7.2
Layer 3. Executable observer			
An observer that records and acts	A finite observer executes repair, retains its records and chooses later interactions from them; feedback changes the outcome law against disconnected and shuffled controls	certified finite computation	§6
Measurement inside an interacting field process	Flux, charged matter and a scalar field on one register with complete instruments and feedback; a certified nonzero infinite-occupation response	certified finite computation	§6.1

Table 2 continued

Structure	What the tower returns	Class	Where
Agreement between overlapping observers	Two observers sharing a port read one complete law of fields, local clocks, feedback and timeouts; native and effective descriptions agree exactly, with a quantum extension and a compiled apparatus within 10^{-11}	certified finite computation	§6
Clocks made of events	Local event clocks with complete stopped instruments; conversion between two clocks keeps their whole count distribution; an active clock whose frequencies are recovered from record covariances	certified finite computation	§6
Layer 4. Fields, clocks and causal order			
A local field operator on the screen	One repair history gives a regular, strongly local Dirichlet form of spectral dimension two, Gaussian density laws $e^{-\lvert\tau-u\rvert A}$ and admitted waves $\cos(t\sqrt{A})$ of finite speed on a declared wave processor	source limit theorem	§7.1
An effective Lorentzian field theory	On a declared radial product, a Lorentz-signature wave density and one smooth static $(1+3)$ metric whose free scalar reproduces every finite menu of observer readouts to any tolerance	conditional analytic theorem	§7.1
Lorentz kinematics and the velocity space	$\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1)$ with H^3 the space of future unit timelike directions	classical identification on the supplied spherical support	§7.2
A source-derived $1+3$ causal precursor	The real axis and the rank-three source quotient give Lorentz inertia $(1,3)$ with the S^2 null boundary, and authenticated logs embed by exact order embeddings	finite theorem plus reconstruction implication	§7.2
Proper time from counting	Record counts in source-record diamonds recover proper-time ratios by the unique fourth-root readout	finite theorem	§7.2
A nonlinear matter continuum and an interacting state space	First-order trajectory convergence for the real quartic-wave sector of a supplied charged action, and a unique self-adjoint closure of its declared Hamiltonian	conditional analytic theorem	[8]
Layer 5. Gauge structure and matter			
The Standard Model gauge Lie type	Complete response with internally implemented rechartings leaves $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ as the unique survivor on twelve ports	finite theorem with declared classification inputs	§8
One anomaly-free generation of fifteen states	Exhaustive scan of the declared exterior menu leaves one charge-conjugate pair of rank-fifteen chiral anomaly-free masks, with common kernel $\mathbb{Z}_6$	finite theorem	§8
Absence of the minimal grand-unified exchange channel	The forced adjoint contains no X/Y generator	finite theorem	§8

Table 2 continued

Structure	What the tower returns	Class	Where
Residual mismatch from holonomy	Fixed nonzero cycle holonomy forces a strictly positive minimum mismatch that vertex repairs cannot remove	finite theorem	§8.2
The relativistic mass shell and internal energy in inertia	$E^2 = m^2 + \|p\|^2$ for the declared-mass momentum, and two declared selection routes giving inertial coefficient $m + E$	finite theorem	[8]
Coulomb and Maxwell structure	Unique minimal-energy neutral Coulomb solution, a local twenty-face curvature action gauge invariant for conserved seam currents, and a hopping charge conserved at a declared nonzero step h	finite theorem	[7]
No explicit photon, gluon or graviton mass term	Three conditional structural statements with zero hard mass parameters on their declared branches; confinement and a quantum graviton lie outside them	reconstruction implication	[11]
Layer 6. Gravity			
The Einstein field equation with Λ	Tensor completion of a realized null balance under Ward conservation, the Bianchi identity, and one scale identification	reconstruction implication	§9
Entropy that scales with a boundary	Collar decomposition with one-sided edge term $\sum_\alpha p_\alpha \log d_\alpha$; the sparse crossing-read functional Γ-converges to perimeter; on the two-dimensional source the native return reference has mutual information at most $\log J + 2025/(64V)$, too small to scale with the crossing seams of a base-face patch	finite theorem; conditional analytic theorem	[9, 85]
Layer 7. Cosmology and tests			
Flat rotation curves and the baryonic Tully–Fisher relation	Under deep scale covariance and quadrature composition the unique law gives $v^4 = GM_b a_0$; a_0 is not source-derived	conditional characterization theorem	§10.1
The dark-energy equation of state	Fixed record capacity under $\rho_{\mathrm{DE}} = \kappa/N$ gives $(w_0, w_a) = (-1, 0)$	reconstruction implication	§10.1
The de Sitter transfer sign	Exact capacity identity with $\Delta S_{\mathrm{gen}}^{\max} = \log(1 - f) < 0$	finite theorem	[13]
The cosmological constant	Capacity candidates 3.2921×10^{122} and 3.3001×10^{122} against 3.3129×10^{122}	diagnostic	§10.2
The fine-structure constant	Interval-certified screen-grain roots 136.994835 and 137.035660 with a declared quark-screening ansatz, and 136.3828, excluding the measured value, with the measured hadronic payload; Wissner-Gross's independently specified entangled-pair scalar has inverse 137.035999189566 [104], with electromagnetic identification unproved	diagnostic; conditional spectral candidate	§10.2
The Koide relation	Positive-chamber circulant identity with $Q = 2/3$ exactly at $\rho/a = 1/\sqrt{2}$, with a custody-bound tau window; the balance premise was abstracted from the measured triple	finite theorem; prospectively fixed conditional test	[12]

Table 2 continued

Structure	What the tower returns	Class	Where
A positive Yang–Mills mass gap	The continuum gap identified with the limiting repair gap under named finite-rate, transfer and reconstruction certificates, for pure Yang–Mills theory with a compact simple group; the transfer premise fails for the $\mathbb{Z}_2$ Wilson transfer matrix at every tested interacting coupling	reconstruction implication	[16]
Isotropy of propagation to angular rank five	The first symmetry-allowed anisotropic rank of the carrier class is six, with two frozen rays inside one exact band	finite theorem; frozen prospective branch prediction	§12.1
Bell correlations beyond local records	Public loophole-free records give $S = 2.42250$ against the bound $S \leq 2$ of the diagonal branch read as a local response model	diagnostic	§12.3
An instrument comparison fixed before data	One coherent event reads the record law's coherence multiplier g; C predicts 5/8 and 1/8, E predicts 3/8, with a complete-count rejection rule	prospectively fixed conditional test	§12

11.2 The status of each layer

Table 3 gives the status of each layer. "Established" means proved on its stated domain or executed with retained receipts; "conditional" means an implication whose displayed antecedents the model does not realize; the open join is what the executable model must supply on the same process; and the stopping test is the result that would remove the layer, its branch or the tested law. Negative results are findings about the tested law, stated with their exact scope.

Table 3: Status of each layer of the tower on the executable model.

Layer	Established and conditional	Open join	Negative results and stopping test
0. Closure and agreement	Established: on the finite consensus branch the closure map is an affine idempotent fixing exactly the packets on consensus normal forms; the record-closure equation is posed exactly. Interpretation and acceptance criterion, not a theorem	A history domain and closure operator for the executable model; interpretation maps fixed independently of validation outcomes, compatible across observers and refinement	The bounded completion class does not identify the solution N; no theorem selects a unique universe. A contradiction among indispensable premises stops the proposal

Table 3 continued

Layer	Established and conditional	Open join	Negative results and stopping test
1. Repair mathematics	Established: canonical observable normalizer; consensus normal form under the transactional diamond; adaptive-run, capacity and fixed-federation bounds; Born weights without continuity axiom, Lüders selection, Tsirelson bound, no-copying; positive residual mismatch from fixed holonomy. Conditional: the finite algebra-state representation of completed records is an input	A source-produced complex-tomographically-complete effect and instrument web with full coexistent-effect additivity; a scheduler selected by the process	No scheduler is selected; deciding consistent extension is NP-complete for succinct Boolean systems; retained routing logs do not identify federation gluing. As theorems they are refuted only by a counterexample
2. Carrier and repair law	Established: the seam rule from four requirements; the classified convex record-law family; the gap $(5-\sqrt{5})/60$ for every admissible table; the rank-three slow-response completion; one seam per native repair event. Conditional: physical position consumes (M1)	One record law and federation gluing selected on the physical process; a physical length	At full-port energy scaling the refining screen loses the rank-three band. The C/E instrument test separates C, E and the current-only law; a resolved dispersion outside the band excludes the whole carrier class
3. Executable observer	Established: native observer with record-caused feedback; electromagnetic and scalar detectors; one interacting process with certified infinite-occupation contrast; overlapping observers exact on 5,768 output words; local event clocks; identification of the record law; coherent observers on matched screens; complete $O(1/d)$ error for the smooth four-read protocol; a continuing C/E finite-buffer observer with counted exports and an exact stationary source/complete-record marginal	One continuing autonomous observer with every service, memory, reset and resource accounted; apparatus, schedules and preparations derived instead of declared	Outcome accuracy does not certify the output state; committed native records form a classical process, so correlations beyond $S = 2$ need declared coherent preparation services
4. Fields and clocks	Established: the native continuum operator of spectral dimension two with Gaussian fields, decay and finite-speed waves; the finite $(1+3)$ causal precursor and count clock. Conditional: the smooth static $(1+3)$ scalar description on a declared radial product; the continuum certificate	A physical clock and energy normalization; an interacting chiral $3+1$ field description from native records; a source-selected radial observation algebra	Passive repair has no wave sector and implements no nontrivial modular automorphism; the native reference identifies no quantum vacuum; the eigenvalue count excludes an elliptic three-manifold identification of the screen operator

Table 3 continued

Layer	Established and conditional	Open join	Negative results and stopping test
5. Gauge and matter	Established: the forced Lie type; the finite response witness; the exterior scan and $\mathbb{Z}_6$ quotient; the residual-mismatch theorem. Conditional: the reversible response D and laboratory currents consume (M3); matter identification consumes (M3) and (M5)	A localized native defect sector with its invariant, maintenance energy, kinetics, charge, spin and statistics in the common clock and energy dictionary; the measured spectrum and mixing	The double-load candidate is transient, surviving with probability $(5/6)^n$; the 66-state binary class has no nonconstant invariant; the finite mass laws give $1 : 1.19 : 1.50$ against $1 : 206.8 : 3477$. An exhaustive no-go over a frozen defect class stops that mechanism
6. Gravity	Established: null tomography and the tensor completion from an explicit null-balance premise. Conditional: the Einstein equation up to Λ under (G1) to (G6)	(G1) to (G6) realized on one refinement tower with the same matter, modular flow and clocks; defect maintenance energy in the same stress	The static radial product fails null balance for its own scalar excitation; stationary repair entropy is distinct from fixed-volume generalized entropy. Failure of (M4) removes the spacetime and Einstein conclusions together
7. Cosmology and tests	Established: the deep-law uniqueness under two premises; $(w_0, w_a) = (-1, 0)$ at fixed capacity; the de Sitter sign; the dispersion band. Diagnostics: capacity and grain roots	Source-only selection of P, N and a_0; physical encoding and calibration for the C/E test	The measured hadronic payload gives a grain root excluding the measured coupling. Exclusion of the band, a C/E rejection, a charged-lepton average beyond three standard uncertainties from the balanced center, or exclusion of $(-1, 0)$ fails its branch

11.3 The realization maps

Five maps name what a physical realization must supply, and each boundary statement in this paper cites them by label.

(M1) Position and scale. An identification of the intrinsic carrier completion with physical position, with a physical length, cofinal overlap gluing between carriers, and a refinement limit (Section 5).

(M2) Clock and energy calibration. An identification of the internal step with a laboratory clock and of the modular ledger with a laboratory energy, in measured units; configuration-derived model duration does not select laboratory time or its units. Native event clocks, an active clock and an inverse clock prediction supply joint clock tests in count parameters, and a direct count-to-phase attachment to published optical ratios is rejected under its readout assumptions (Section 12.3).

(M3) Laboratory current. An identification of the finite port response with a physical gauge current, with a physical matter action and a continuum operator limit.

(M4) One common physical tower. A single physical refinement family realizing the event, modular, stress, entropy, continuum, vacuum, and scale premises together, rather than several independently declared islands.

(M5) A source-selected action. An action produced by the source rather than declared, with its field sector, coupled interaction, relative normalizations, and operational readout.

The executable observer turns these maps into construction work on one process, and the open joins of Table 3 are their instances. A finite theorem that consumes none of the five maps stands on its own domain, and no result in this paper discharges any of them; Section 12.3 tests two simple attachments of native records to published data. The construction paper gives the detailed premise map, including the retained negative controls that bound how the carrier, the response triplets, the feedback class and the kinetic term may be attached [66].

12 Empirical meaning and falsifiability

The custody-bound register fixes target definitions, exclusion thresholds, precision floors, exposure classes and decision rules before eligible comparison [3]. Two kinds of test follow from the construction: a propagation anisotropy fixed by the carrier's symmetry, and an instrument comparison fixed by the executable observer. Published records also test precise attachments of native records retrospectively.

12.1 Propagation anisotropy at angular rank six

Angular rank means spherical-harmonic rank j, not a particle spin. The dimension of the invariant subspace of the sixty proper icosahedral rotations in angular rank j is

$$m_j = \frac{1}{60}\Big[(2j+1) + 15\,U_{2j}(0) + 20\,U_{2j}(\tfrac{1}{2}) + 12\,U_{2j}(\tfrac{\varphi}{2}) + 12\,U_{2j}(\tfrac{\varphi-1}{2})\Big],$$

with U the Chebyshev polynomials of the second kind, and the exact table gives $m_1 = \cdots = m_5 = 0$ and $m_6 = 1$ [49].

Theorem 12.1 (Icosahedral angular-rank universality)**.** *Let the carrier kinetics be any operator $\lambda_a(k) = a^{-2}\sum_d w(d)\,[1-\cos(a\,k\cdot d)]$ with a finite direction multiset and weights invariant under the proper icosahedral rotation group. Then the dispersion is exactly isotropic through angular rank five at every order in a, every directional term below angular rank ten is one multiple of the unique normalized invariant $\mathcal{I}_6$, and whenever the weighted sixth-moment coefficient is nonzero the first directional artifact appears at order a^4k^6, suppressed by exactly $1/16$ under one binary refinement at that order. Rank six is the least symmetry-allowed nonzero anisotropic rank in the class.*

Proof. The order-$a^{2m-2}k^{2m}$ angular content is an invariant polynomial of degree $2m$ with harmonic components at even ranks $j \le 2m$, and the invariant table forces zero at $L = 2, 4, 8$ and a one-dimensional space at $L = 6$. The table, the 62-direction stationary census, the orbit verification and the nonvanishing sixth-moment coefficients are certified in exact arithmetic [1]. □

Proposition 12.2 (Primitive twelve-port branch prediction)**.** *Suppose the intrinsic spatial kinetic symbol lies in the cosine class of Theorem 12.1 with scale-independent coefficients, the complete orbit of the twelve port directions u_i is its sole hop support through order k^6, the carrier scale a is finite and positive with the quadratic term normalized to k^2, one carrier rest frame is transported coherently over the experiment, and the tested scalar or polarization-independent sector realizes the operator. Then*

$$\omega^2 = \frac{1}{2a^2}\sum_{i=1}^{12}[1-\cos(ak\,u_i\cdot\widehat{k})] = k^2 - \frac{a^2}{20}k^4 + \frac{a^4}{840}k^6 + \frac{2a^4}{7875}k^6\mathcal{I}_6(\widehat{k}) + O(a^6k^8), \tag{14}$$

with $\mathcal{I}_6(n) = \frac{25}{132}\sum_i P_6(u_i\cdot n)$*, so in a transported frame the branch predicts the scale-free relations*

$$\frac{B_6}{C_4^2} = \frac{32}{315}, \qquad \frac{B_0}{C_4^2} = \frac{10}{21}, \qquad \frac{B_6}{B_0} = \frac{16}{75}, \tag{15}$$

and every intrinsic anisotropic coefficient at $1 \le j \le 5$ *vanishes.*

Proof. The exact carrier moments $\sum_i (u_i\cdot n)^2 = 4$, $\sum_i (u_i\cdot n)^4 = 12/5$ and $\sum_i (u_i\cdot n)^6 = 12/7 + (64/175)\,\mathcal{I}_6(n)$ substituted into the cosine series give the expansion; eliminating a gives the ratios. □

A second frozen ray follows from the source seams.

Proposition 12.3 (Source-seam edge ray and conditional propagation branch)**.** *The boundary current of the complete thirty-seam source packet maps through the antipodal-odd load readout onto the even-sum module* D_6 *of Theorem 5.1, with Smith invariants* $(1,1,1,1,1,2)$*, and every directed seam defines one translation of the internal record carrier whose complete normalized direction multiset is the thirty-direction edge orbit. If the feasible move laws and their objective are natural under the proper carrier action and A3 supplies a unique normalized minimizer, transitivity forces weight* $1/60$ *on every directed seam, and the selected internal operator is the source-counting homogeneous convolution.*

Suppose these record translations are physical displacements under one homogeneous position action, that the complete edge orbit is the sole direct support through the displayed order, that proper-carrier covariance fixes one common weight, that the same operator acts on one scalar or polarization-independent physical sector with quadratic term normalized to k^2*, and that the action carries the stated gluing, scale, frame, readout, nuisance, and exclusivity data. Its spatial kinetic symbol is then*

$$\Lambda_a(k,\widehat{k}) = \frac{1}{5a^2}\sum_{j=1}^{30}\Big[1-\cos\Big(ak\,w_j\cdot\widehat{k}\Big)\Big], \tag{16}$$

with $C_4 = -a^2/20$*,* $B_0 = a^4/840$ *and* $B_6 = -a^4/12600$*, hence* $B_0/C_4^2 = 10/21$*,* $B_6/C_4^2 = -2/63$ *and* $B_6/B_0 = -1/15$*. The intrinsic anisotropic coefficients at angular ranks one through five vanish, and the rank-six vector lies on the rotated* $\mathcal{I}_6$ *orbit.*

Proof. Exact integer incidence gives the displayed image and Smith invariants, and the response-selected Gram pullback gives the completion of Theorem 5.1. The sixty directed labels map two-to-one onto the thirty signed edge directions, whose unit-normalized moments are $\sum_j (w_j\cdot n)^2 = 10$, $\sum_j (w_j\cdot n)^4 = 6$ and $\sum_j (w_j\cdot n)^6 = 30/7 - (2/7)\,\mathcal{I}_6(n)$; substitution in (16) gives the coefficients and ratios. The incidence, quotient, orbit binding and moment identities are machine-checked in exact arithmetic [1]. □

The proposition contains an exact finite source theorem and a conditional physical branch. The naturality and minimizer clause supplies a homogeneous internal action whose generator satisfies the positive maximum principle and an exact Dirichlet identity, and whose plane-wave spectrum is the symbol above. The declared symbol also carries an exact remainder certificate with no physical premise: with $q = ak$, $|\widehat{\Lambda} - P_6| \le \frac{7}{388800}q^8$ and $\frac{19}{20}q^2 \le \widehat{\Lambda} \le q^2$ uniformly on $0 \le q \le 1$, where $P_6 = q^2 - q^4/20 + (1/840 - \mathcal{I}_6/12600)q^6$; the certificate turns a into no SI length. The edge branch and the primitive-port branch have opposite rank-six signs, and their coefficient rays, custody packages, exposure exclusions and comparison budgets are separate.

Theorem 12.4 (Carrier-class dispersion band)**.** *For the declared positive-weight scalar cosine class over orbits O_s of the sixty proper carrier rotations with radii $r_s > 0$, weights $w_s > 0$ and the k^2 coefficient normalized to one, write $\Lambda_{\mathrm{norm}} = k^2 + C_4 k^4 + (B_0 + B_6 \mathcal{I}_6(\widehat{k}))k^6 + O(k^8)$ and $\mu_m = \sum_s w_s |O_s| r_s^m$. Then $C_4 = -(a^2/20)\mu_4/\mu_2 < 0$ for every member; $B_0/C_4^2 = (10/21)\mu_2\mu_6/\mu_4^2 \geq 10/21$ with equality exactly on single-radius members; the anisotropic ranks one through five vanish; and B_6/B_0 lies in $[-16/135, 16/75]$, the face and vertex orbits attaining the endpoints and the edge orbit sitting at $-1/15$.*

Proof. Summing over the sixty rotations gives $\sum_g ((gu)\cdot\widehat{k})^6 = \frac{60}{7} + \frac{64}{35}\mathcal{I}_6(\widehat{u})\mathcal{I}_6(\widehat{k})$ on unit vectors. The floor is the Lagrange identity $\mu_2\mu_6 - \mu_4^2 = \sum_{i<j} W_i W_j r_i^2 r_j^2 (r_i^2 - r_j^2)^2 \geq 0$ with $W_s = w_s|O_s|$, and the band holds because the rank-six numerator is a positive-weighted mean of seed values of $\mathcal{I}_6$ in $[-5/9, 1]$ [1]. □

A resolved intrinsic dispersion with B_0/C_4^2 below $10/21$, or B_6/B_0 outside $[-16/135, 16/75]$, or a residue at ranks one through five, excludes every member of the declared class at once. The primitive-port relation is a frozen prospective conditional physical-branch prediction: its target statement, coefficient snapshot, registration record, public commit custody and detached calendar attestations fix the prospective content, the dated template search whose icosahedral templates returned a family-wide p-value of 0.64 is ineligible, and a verdict requires a dataset-specific registration with a joint likelihood, frames, orientation profile, nuisance model and calibrated threshold. The branch fails at five standard deviations or more if an isolated intrinsic C_4 is positive, a lower-rank coefficient is nonzero, or the calibrated profile likelihood excludes the complete manifold $C_4 < 0$, $B_0 = \frac{10}{21}C_4^2$, $B_6 = \frac{32}{315}C_4^2$. A null is inconclusive, since the branch supplies no positive lower bound on a, and a failure rejects the primitive twelve-port propagation branch; it reaches the framework as a whole only after a stronger source law makes that branch forced and exclusive [3].

12.2 An instrument test from the executable observer

Under a common current-record interface every admissible record law has the classical statistics of C and E. The multiplier g, the coherence between an empty seam and a waiting single load, separates C, E and the law that records only the current. On a fixed coherent two-state source preparation the readout probability is $3/8 + \frac{1}{2}\,\mathrm{Re}(g\,e^{-i\theta})$: reversing the phase moves C from 5/8 to 1/8, E predicts 3/8 for both, and the law that records only the current predicts $3/8 \pm \sqrt{2}/4$ [68]. Four fixed settings include common-phase and incoherent controls. The complete-count evaluator uses 8192 trials per setting, an independently established probability-error budget of 1/64 and an exact sampling certificate below 1/1000; it rejects malformed or incomplete records and reports *no data* when no independent counts are supplied. A source-executed implementation runs both complete 78-state instruments, the four preparations and a complete Lüders readout from twelve-port control words, retaining every record and attempted trial [66]. A sequential design on one continuing carrier needs neither exact resets nor independent trials: 379 discarded native attempts before each of 3,927 trials replace the reset, failed heralds count as trials, and conditional-law bounds on the dependent transcript give a wrong-candidate probability of at most 3/400 under the exact-source model [87]. Synthetic controls exercise the decision rules and establish no physical verdict; physical encoding, independent calibration and comparison data are the open requirements.

12.3 Native records against public data

Published records can test precise attachments of native records without a new laboratory realization; each test below is retrospective, states its readout hypotheses, and assigns no prospective

verdict.

The diagonal native branch is a classical stochastic process, so under independently chosen settings and local factorized responses it obeys the CHSH bound $S \leq 2$. The deposited records of the loophole-free Hensen experiment [90] reproduce 245 eligible trials, 196 wins and $S = 2.42250$, and the published fixed-count analysis gives a tail bound near 0.03917 with its setting-calibration allowance [88]. This rejects the local classical reading of the diagonal branch under that experimental contract. Committed native records form a classical process: the native grammar keeps diagonal inputs diagonal under every classical feedback history, and two implementations with the same committed branches, which give 3/4 and 1/4 under a read before commitment, give 1/2 for both once the record is committed [98]. Correlations beyond $S = 2$ therefore need uncommitted coherence from declared preparation services, and every admissible record law attenuates it [68]. The replay leaves the completed record surface untouched, on which a declared entangled state reaches the Tsirelson bound of Theorem 4.5, and it does not distinguish C from E.

Event clocks need no external validation, but identifying their readings with a published measurement is a testable hypothesis. Let two native ports, or two distinct regions of one carrier, count incident repairs, and attach the published 300-second blocks of the Ar^{13+}/Yb^{+} optical frequency ratios of King et al. [91] to the normalized count ratio through a gain of at least one, a declared bracket on the Yb^{+} frequency and an absolute error of 10^{-12} per block, with direct phase-count readback. The complete count law, after charging every choice of 2021 subwindow, gain and retained blocks, bounds the probability of the observed block ranges below 10^{-104} for two native ports and below 10^{-22} for every pair of the 3,433 distinct regional event sets [89]. The direct attachment is rejected together with its readout assumptions; the experiment's interrogations and servos lie outside that hypothesis, and the native event clocks are unaffected as clocks of the model.

Published trapped-ion records test a supplied experimental model executed through the source interface. Two calibrated preparation parameters predict 100 reserved complex readouts with per-spin-mean root-mean-square error 0.06647, and a changed-potential transfer predicts 125 derived observations with root-mean-square error 0.2876 in position-estimator units [67]. These comparisons test the supplied oscillator model and derive neither its action nor its apparatus from native repair.

12.4 Diagnostics and what fails together

The other quantitative comparisons carry their own decision shapes and do not count as predictions. The closure roots of Section 10.2 are diagnostics of declared maps. The registered conditional primordial branch $n_s = 1 - P/48 = 0.96602$, at the calibrated grain P, lies at the lower 0.612 percent position of the official ACT DR6, Planck and DESI joint posterior chains, whose mean is 0.9743 ± 0.0034 [92, 89]. Those data precede the registration, so this is a retrospective tension under baseline ΛCDM, not a likelihood ratio, and the registered comparison with a later dataset is unchanged. A separate ACT DR6 likelihood slice on the measured bandpowers, with the background and absolute calibration held fixed, prefers a free tilt, fitted at 0.9773, by $\Delta\chi^2 = 4.07$; no significance or framework-wide verdict is assigned to it [89]. A charged-lepton world average more than three standard uncertainties from the balanced-circulant center refutes the equal-rank-two tracial reading of the face-circulant sector; a resolved deep-regime galaxy sample violating degree-one radial scale covariance or quadrature composition removes the uniqueness of the deep law; and exclusion of $(w_0, w_a) = (-1, 0)$ fails the fixed-capacity branch. The conditional implications share premises, so they fail in groups: failure of (M4) removes the effective-spacetime and Einstein conclusions together, failure of (M3) removes the gauge and matter interpretations together, and failure of (M2) removes every statement in laboratory units. The finite normal-form, invariant-theory and

algebraic results retain their stated domains under all three.

13 Machine verification and reproducibility

Each result maps to an artifact class: a Lean library with no admitted propositions in the dependency closure reported here, including formal premise boundaries and countermodels; exact-arithmetic code receipts, integer and rational computations and outward-rounded interval certificates, each emitting a canonical hashed payload; executable observer receipts that replay every retained result and check its artifact hashes; simulation receipt bundles pinning the source revision, configuration, seed where stochastic, tolerances and output hashes; verifiers that recompute verdicts from clause vectors and reject receipts whose stored verdict disagrees; and adversarial negative controls, so that every certificate ships with mutations that must be detected. Machine-readable records classify every quantitative row as a source result, reconstruction implication, diagnostic, prospectively fixed test or rejected candidate, and reject any stronger classification unsupported by the recorded dependencies.

Table 4 names the artifact family behind each group of results. The repository, receipts, schemas and rebuild instructions are public [1, 2, 67]. The formal and executable stack checks the claims assigned to it and enforces their stated boundaries; analytic arguments in this manuscript retain their displayed proofs, and no part of the stack establishes physical realization.

Table 4: Principal verification artifacts and their scope.

Results	**Principal artifacts**
Executable observer, detectors, interacting process and C/E instruments	The observer receipts with exhaustive path replays, seeded executions, feedback controls, exact and interval certificates, and a single verification entry point that replays every receipt and checks its hashes [67, 66]
Overlapping observers, local clocks, screens and the continuum operator	The model packages for the observer correspondence, the compiled clock history, joint and active clocks, repair identification, federated screens and the coherent screen observer, each with exact or interval receipts, independent oracles and replay; the analytic limit arguments with their constants and order of limits [76, 83]
Public-data comparisons	Hash-pinned public records (Bell trials, optical ratio traces, CMB posterior chains) with exact rational reducers, their stated readout hypotheses and replay checks [88, 89]
Normal forms, stability, refinement, complexity and the transactional diamond	The observable-normal-forms library and component-paper reductions; the local-diamond and Newman arguments of this manuscript; the finite transaction-model verifier
Born weights, Lüders selection, Tsirelson saturation, no-cloning	The event-algebra modules for the interface, conditioning, the bound and the public record algebra, with per-module axiom audits
Rank-three completion	The port-Gram repair band, covariance and frame-quotient modules with a source-pinned receipt and independent verifier
Gauge Lie type, matrix witness, matter image and exterior selection	The screen modules for the carrier action, commutant and incidence response; the exact-arithmetic port current certificate; the trace-balanced kernel and exterior scan modules
Causal precursor, count limit and proper-time readout	The provenance and geometry modules; the source-record family receipts with exact all-pairs counts through $q = 89$ and the sampled $q = 144$ level, the carrier realization receipts and event logs, and the public archive

Table 4 continued

Results	Principal artifacts
Null tomography, small-ball arithmetic and tensor completion	The Einstein-branch tensor module for null directions, design, decoder, injectivity and metric ambiguity; explicit null-balance, Ward, Bianchi and scale premises
Angular-rank theorem, frozen branches and the dispersion band	The primitive-port prediction and carrier-class band modules with mutation tests; the custody-bound prediction register
Exact federation and the closure loop	The level-six configuration, seeds, seam arrays, loads, per-schedule ledgers, quotient hashes, standalone verifier and public archive; the level-ten settlement receipts; the event logs and recovered specifications of the canonical and control sources
Dark sector, vacuum coordinate and closure roots	The dark-sector, deep-profile, per-cut and fixed-capacity modules with the matched seen-data diagnostic; the capacity closure and nonidentifiability modules; the interval-certified grain root

14 Discussion and comparison

Operational reconstructions derive quantum structure from information and composition principles [17, 18, 19]. OPH makes a narrower statement at the quantum stage: it takes the finite algebra-state representation as an input and contributes the completed-record surface on which the event, conditioning and correlation identities operate, together with the consensus theorem that defines completion. Quantum Darwinism locates classical objectivity in redundant environmental records [44], and the completed-consensus record plays the same public role with redundancy replaced by protected overlap agreement and an explicit repair dynamics. The cellular-automaton programme shares the finite substrate [45], while OPH locates the quantum identities on the completed record surface. Wigner's-friend extensions constrain agents reasoning about one another's unfinished measurements [46], whose premises concern pre-consensus perspectives that the completed surface does not represent. Relational quantum mechanics treats physical facts as relational [42] and the thermal-time programme reads time from modular structure [43]; OPH attaches public facts to protected overlap records, and its executable observer reads modular flow from the actual observation state of a running process and lets native incident events drive a modular step whose generator is the native transfer, with a faithful thermal reference derived from closed native histories; passive repair alone implements no nontrivial modular automorphism. The stationary native state of a fixed-number sector is tracial, so its modular flow is trivial, while conditioning on a native current event gives a Palm reference whose coherent modes carry modular frequency $\log(5/4)$, the ratio of the ten moving branches of a non-adjacent occupied pair to the eight of an adjacent one, a degree-census identity of the carrier: modular time in this construction is relative to an event [67]. The local-diamond proof sits beside asynchronous agreement [38, 39], and Dijkstra's self-stabilization separates recovery from continued legitimate operation [40].

The gravitational branch belongs to the thermodynamic and entanglement-based routes to geometry [53, 54, 57], with Bisognano–Wichmann modular flow and half-sided modular inclusions as established analytic ingredients [50, 51, 52]. The route from a null-projected balance and the Bianchi identity to the full equation, with Λ as an integration constant, is familiar from trace-free and unimodular formulations [55, 56]; what is added is the finite nine-direction design with exact constants and the premise contract (G1)–(G6) stated as one implication. The gauge section uses Doplicher–Roberts reconstruction for the sector group [59, 60] and contributes the forcing of the local Lie type from complete response and rechartings inner on that response, a stronger premise

than mere internal availability of their implementers.

The closure hypothesis has precedents of a different kind. Wheeler pictured a participatory universe as a self-excited circuit [100]. Tegmark's mathematical-universe hypothesis identifies physical existence with mathematical structure [101], and Nozick proposed that a fundamental principle might subsume itself [102]. OPH adds two demands: the self-subsumption must be inhabited, carried out through records and agents inside the structure, and it must be testable in part, through an executable specification whose observers can be run and whose instruments make predictions. Lawvere's fixed-point theorem shows why unrestricted self-representation fails [103]; the closure condition is restricted to the observable and interventional structure that observers can recover.

A useful comparison of unification proposals asks how much physical structure follows from each declared input and which observations distinguish the resulting models; OPH offers a concrete case with proofs and executable checks attached to individual implications. Formal verification measures deductive reliability, predictive success requires physical realization and eligible measurements, and neither theorem counts nor numerical coincidences define a probability that an architecture describes nature.

15 Conclusion

The goal is a consistent and complete theory: the closure hypothesis realized as one executable process whose observers measure what we measure, with every layer of the tower carried by the same law, state, clocks and refinement, or else proved obstructions that remove OPH or a stated branch of it. This paper places the evidence on that tower. Layer by layer the finite theorems turn readouts of the executable model into familiar structure: public records independent of the repair schedule, the Born rule and the Tsirelson bound on completed records, a carrier whose slow response has exactly three directions, the Standard Model gauge Lie algebra with a conditional anomaly-free generation, a source-derived $(1 + 3)$ causal precursor with proper-time ratios from record counts, and the Einstein equation conditional on independent null balance and conservation. The model runs: its observer acts on retained records, one interacting quantum process carries flux, charged matter and a scalar field through complete measurements, overlapping observers read one complete law, and joined carriers on a refining screen carry one continuum operator whose effective description on a declared radial product is a smooth $(1 + 3)$ Lorentzian field theory.

The open joins are fixed by the hypothesis itself. One process must supply a physical clock and energy normalization, an interacting chiral $3 + 1$ field description of native records, a localized defect sector whose maintenance energy is its rest energy, and a responsive geometry realizing the Einstein premises. The negative results bound where these joins can be found: the double-load defect is transient, passive repair has no wave sector, the static radial product fails null balance, and the finite mass laws miss the charged hierarchy. The five realization maps are consumed rather than constructed, so these results do not establish that the architecture describes our universe. The prospective tests are the frozen dispersion band of Theorem 12.4 and Proposition 12.2, under which a resolved intrinsic dispersion with $C_4 \geq 0$, with $B_0/C_4^2 < 10/21$, with B_6/B_0 outside $[-16/135, 16/75]$ or with a nonzero coefficient at angular ranks one through five excludes every member of the declared carrier class at once, and the C/E instrument test, whose decision rule is fixed before any independent count.

Reproducibility statement

All theorems, certificates, simulation configurations, receipts, and the formal library are public in the project repositories [1, 2]. The paper release manifest records the release identifier (r2055) with PDF hashes and sizes; computational receipt bundles pin the applicable source revision, configuration, seed and output hashes, with clean-checkout rebuild instructions and negative controls; and machine-readable claim and prediction provenance accompanies the artifacts [3]. This manuscript synthesizes and strengthens results whose longer proofs also appear in public component preprints on observable normal forms, finite event algebras, consensus, Einstein reconstruction, gauge structure and the executable observer [4, 5, 6, 9, 11, 66], a relationship disclosed here so that overlap is visible.

Statements and Declarations

Funding. The authors declare that no external grants or dedicated third-party research funding were received for the preparation of this manuscript.

Author contributions. Bernhard Mueller developed the OPH concept and mathematical synthesis, wrote the main manuscript, and coordinated the formal and computational evidence. Mario Poneder developed the executable observer construction, including the native observer, the interacting source process and their certificates, and contributed the finite port-current construction and verifier, gauge and matter certificates, and proof review of the edge-entropy and Einstein-normalization branches. Alexander Osika proposed that the screen pixel ratio and the correctable-record capacity are derivable rather than fitted, supplied the detuning fixed-point equation that determines the pixel ratio, and originated the echosahedral patch-hardware design on which the screen-microphysics branch is built, in addition to the physical-realization and hardware boundary, prototype framing, and manuscript review. Kai Xue contributed consistency checks, simulator review, visualization review, and prototype-engineering review. Ben Cassie contributed implementation-claim and physical-evidence review. Peter Nguyen contributed proof auditing, branch-boundary analysis, compact-gauge and global-quotient analysis, and manuscript consistency review. Jinwook Kim developed the constructions, proofs, and counterexamples for the profile partition and structural characterizations, exact behavior-cut recovery, exact finite-state profile transport, adaptive repair stratification, the cumulative attempt-capacity classification, fixed computation-federation progress, and the fixed-federation execution classification, together with the mechanism-variant comparison kernel and conditional example and the source-derived phase-effect selection. He constructed a provably fair emission-order round-robin scheduler with a linear exact-output attempt bound, established order-sharp quadratic accepted-step bounds for the actual compiler with the exponential fanout control, and proved the historical weak-relation fair-stuttering no-go and the weak-fair delay versus bounded-waste horizon separation. He formalized these results in Lean, connected the repair and execution results to the Tower library, and contributed their manuscript, claim registry, ledger and build records. David Matscheko contributed proof review of consensus repair, observable normal forms, modular and Einstein algebra, edge entropy, hypercharge, and the $\mathbb{Z}_6$ quotient. Daniel Ari Friedman contributed independent red-team audits of the papers, the book, the claim and premise ledgers, the Lean library, continuous integration and the custody records, which led to scoped headline claims, displayed values matched to their certified receipts and repaired falsification rules, and the independently reviewed programme-obstruction audit whose controls bound how the carrier, the response triplets, the feedback class and the kinetic term may be attached. Jonathan Hill contributed Lean formalization and proof auditing of observable nor-

mal forms, refinement and repair results, complexity classifications, and artifact coverage. William T. Glynn contributed simulator development, computational implementation, reproducible-build infrastructure, and release-validation work in the main OPH research repository. Maarten Antonie Visser contributed consensus-protocol review, physics feedback on the foundations and emergent-spacetime framing, and the source-derived causal-poset reading of authenticated provenance that the event-precedence layer implements. Kale Arnav Anirudha contributed consensus-paper material, asynchronous-agreement analysis, repair-map definitions, and formal review of the protocol surface. Brieuc de La Fournière contributed finite Lean constructions on the source-derived causal-order and carrier-symmetry surfaces, including the cover-relation provenance compiler and the antipodal port-action identification, and external comparison controls for the charged-lepton branch. All authors reviewed and approved the manuscript. All authors agree to be accountable for all aspects of the work.

Corresponding author. Correspondence should be addressed to Bernhard Mueller.
Email: `bernhard@floatingpragma.ai`.

Data availability. The finite witnesses, simulation configurations, receipts, scientific claim registries, and release manifests supporting this study are available in the public OPH and simulation repositories [1, 2]. Individual receipts and their bundle documentation provide the source and output provenance for computational claims. The artifact map identifies the level-six exact-federation archive and its level-five to level-ten extension, the closure-loop archive, and the causal-poset archive.

Code availability. The Lean sources, exact-arithmetic verifiers, finite scans, simulation code, tests, and rebuild instructions are available in the same public repositories [1, 2]. The computational claims in this manuscript identify their principal files in Section 13.

Ethics approval and consent to participate. Not applicable. The study involved no human participants, human data, or animals.

Consent for publication. Not applicable.

Competing interests. The authors participate in OPH-related research, software, simulation, or prototype-development programs and may receive professional or reputational benefit from this work. These relationships are disclosed. The authors declare no other competing interests.

Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- Icosahedral harmonics, finite-difference dispersion, and nonminimal Standard-Model Extension coefficient searches [112, 114, 110, 111].
- Root lattices, graph incidence, transverse projectors, free-vector oscillators, icosahedral edge harmonics, finite-difference symbols, Lorentz-violating threshold kinematics, and nonminimal Standard-Model Extension coefficient searches [106, 108, 115, 113, 109, 110, 111].
- Spherical-design moment theory, lattice dispersion expansions, and Lorentz-violation coefficient parametrizations [107, 113, 110, 105].

References

[1] FloatingPragma (2026). Observer Patch Holography: Lean library, exact code receipts, finite certificates, and scientific claim registries. `https://github.com/FloatingPragma/observer-patch-holography`

[2] Mueller, B. (2026). OPH physics simulator and reproduction source. `https://github.com/muellerberndt/oph-physics-sim`. Exact-federation archive: `https://github.com/FloatingPragma/observer-patch-holography/tree/r2055/evidence/exact_federation_L6_canonical_20260909`. Closure-loop archive: `https://github.com/FloatingPragma/observer-patch-holography/tree/r2055/evidence/closure_loop`.

Causal-poset archive: https://github.com/FloatingPragma/observer-patch-holography/tree/r2055/evidence/source_net_causal_poset. Active refinement-readout revision: https://github.com/muellerberndt/oph-physics-sim/tree/036b12b608a9ca4c19d69f5e951b3b83a7a51cd8.

[3] FloatingPragma (2026). Precommitted comparison certificates, custody records, and the frozen prediction register. Immutable source receipt: https://github.com/FloatingPragma/observer-patch-holography/blob/66176656dc1143f9ec50ba1a6e409c403545857f/code/a5_fingerprint/runtime/spin_six_primitive_port_prediction_receipt.json. Append-only custody packet: https://github.com/FloatingPragma/observer-patch-holography/tree/bb73f80d459113ccc82d5a5b5219d1dda78e9a79/evidence/custody/falsification/frozen_targets/fz11_2026-07-31. Source-seam edge receipt: https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/code/a5_fingerprint/runtime/seam_current_edge_prediction_receipt.json. Source-seam custody and decision-rule clarification: https://github.com/FloatingPragma/observer-patch-holography/tree/bb73f80d459113ccc82d5a5b5219d1dda78e9a79/evidence/custody/falsification/frozen_targets/fz12_2026-08-02. Koide custody packet: https://github.com/FloatingPragma/observer-patch-holography/tree/bb73f80d459113ccc82d5a5b5219d1dda78e9a79/evidence/custody/falsification/frozen_targets/fz10_2026-07-28. Live register: https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/claims/frozen_prediction_register.json.

[4] Mueller, B., Kim, J., Matscheko, D., and Hill, J. (2026). *Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems*. Public manuscript in the same repository revision as this paper. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/extra/observable_normal_forms.pdf

[5] Mueller, B. (2026). *Verified Projection-Event Calculus in Lean 4: Bundled Arbitrary-Partition Pinching, Lüders Retractions, and CHSH Interoperability*. Public manuscript and formal artifact. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/extra/machine_checked_finite_event_algebras.pdf

[6] Mueller, B., Xue, K., Kim, J., Anirudha, K. A., Matscheko, D., and Hill, J. (2026). *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/reality_as_consensus_protocol.pdf

[7] Mueller, B., Osika, A., Xue, K., Cassie, B., de La Fournière, B., and Friedman, D. A. (2026). *Twelve-Port Icosahedral Carriers and Native Seam Repair: The Carrier Layer of Observer-Patch Holography*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/screen_microphysics_and_observer_synchronization.pdf

[8] Mueller, B., Poneder, M., Osika, A., Xue, K., Cassie, B., Nguyen, P., and Friedman, D. A. (2026). *Observers Are All You Need*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/observers_are_all_you_need.pdf

[9] Mueller, B., Poneder, M., Osika, A., Xue, K., Nguyen, P., Visser, M. A., Matscheko, D., and de La Fournière, B. (2026). *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.pdf

[10] Mueller, B., Osika, A., and Friedman, D. A. (2026). *The Fine-Structure Constant as an OPH Pixel Fixed Point*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/extra/fine_structure_constant_derivation.pdf

[11] Mueller, B., Poneder, M., Osika, A., Xue, K., Nguyen, P., Matscheko, D., and Friedman, D. A. (2026). *Standard Model Gauge Structure under Complete-Response and Observer-Transport Premises*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/deriving_standard_model_gauge_structure_from_observer_overlap_consistency.pdf

[12] Mueller, B. (2026). *The Positive-Chamber Koide Identity for Icosahedral Face Circulants*. Public manuscript and formal artifact. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/extra/koide_identity_from_positive_c3_face_circulants.pdf

[13] Mueller, B. (2026). *The de Sitter Time-Advance Sign from a Finite Screen with Fixed Capacity*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/extra/de_sitter_time_advance_sign_from_fixed_screen_capacity.pdf

[14] Mueller, B. and Matscheko, D. (2026). *Observer-Patch Holography and the Dark Sector: Modular Charge, the Anomalous Collar Source, and the Deep Galaxy Law*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/cosmology/oph_dark_matter_paper.pdf

[15] Mueller, B. (2026). *Observer-Patch Holography Cosmological Vacuum and Structure Formation: Finite Transfer Conditions, Fluctuation Ensembles, and Proto-Objects*. Public manuscript. https://github.com/FloatingPra

gma/observer-patch-holography/blob/r2055/cosmology/oph_cosmological_vacuum_and_structure_formation.pdf

[16] Mueller, B. and Hill, J. (2026). *A Conditional Identification of the Yang–Mills Gap with a Finite Repair Gap: Dobrushin Repair Floors and a $\mathbb{Z}_2$ Test of the Transfer Receipt.* Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/extra/yang_mills_gap_clay_problem.pdf

[17] Hardy, L. (2001). "Quantum theory from five reasonable axioms." arXiv:quant-ph/0101012. https://arxiv.org/abs/quant-ph/0101012

[18] Chiribella, G., D'Ariano, G. M., and Perinotti, P. (2011). "Informational derivation of quantum theory." *Physical Review A* **84**, 012311. https://doi.org/10.1103/PhysRevA.84.012311

[19] Masanes, L., and Müller, M. P. (2011). "A derivation of quantum theory from physical requirements." *New Journal of Physics* **13**, 063001. https://doi.org/10.1088/1367-2630/13/6/063001

[20] Gleason, A. M. (1957). "Measures on the closed subspaces of a Hilbert space." *Journal of Mathematics and Mechanics* **6**, 885–893.

[21] Busch, P. (2003). "Quantum states and generalized observables: A simple proof of Gleason's theorem." *Physical Review Letters* **91**, 120403. https://doi.org/10.1103/PhysRevLett.91.120403

[22] Clauser, J. F., Horne, M. A., Shimony, A., and Holt, R. A. (1969). "Proposed experiment to test local hidden-variable theories." *Physical Review Letters* **23**, 880–884. https://doi.org/10.1103/PhysRevLett.23.880

[23] Cirel'son, B. S. (1980). "Quantum generalizations of Bell's inequality." *Letters in Mathematical Physics* **4**, 93–100. https://doi.org/10.1007/BF00417500

[24] Landau, L. J. (1987). "On the violation of Bell's inequality in quantum theory." *Physics Letters A* **120**, 54–56. https://doi.org/10.1016/0375-9601(87)90075-2

[25] Wootters, W. K., and Zurek, W. H. (1982). "A single quantum cannot be cloned." *Nature* **299**, 802–803. https://doi.org/10.1038/299802a0

[26] Dieks, D. (1982). "Communication by EPR devices." *Physics Letters A* **92**, 271–272. https://doi.org/10.1016/0375-9601(82)90084-6

[27] Lüders, G. (1950). "Über die Zustandsänderung durch den Meßprozeß." *Annalen der Physik* **443**, 322–328. https://doi.org/10.1002/andp.19504430510

[28] Bombelli, L., Lee, J., Meyer, D., and Sorkin, R. D. (1987). "Space-time as a causal set." *Physical Review Letters* **59**, 521–524. https://doi.org/10.1103/PhysRevLett.59.521

[29] Surya, S. (2019). "The causal set approach to quantum gravity." *Living Reviews in Relativity* **22**, 5. https://doi.org/10.1007/s41114-019-0023-1

[30] Major, S., Rideout, D., and Surya, S. (2007). "On recovering continuum topology from a causal set." *Journal of Mathematical Physics* **48**, 032501. https://doi.org/10.1063/1.2435599

[31] Myrheim, J. (1978). "Statistical geometry." CERN preprint TH-2538.

[32] Meyer, D. A. (1988). "The Dimension of Causal Sets." PhD thesis, Massachusetts Institute of Technology.

[33] Benincasa, D. M. T., and Dowker, F. (2010). "The scalar curvature of a causal set." *Physical Review Letters* **104**, 181301. https://doi.org/10.1103/PhysRevLett.104.181301

[34] Dowker, F., and Glaser, L. (2013). "Causal set d'Alembertians for various dimensions." *Classical and Quantum Gravity* **30**, 195016. https://doi.org/10.1088/0264-9381/30/19/195016

[35] Hawking, S. W., King, A. R., and McCarthy, P. J. (1976). "A new topology for curved space–time which incorporates the causal, differential, and conformal structures." *Journal of Mathematical Physics* **17**, 174–181. https://doi.org/10.1063/1.522874

[36] Malament, D. B. (1977). "The class of continuous timelike curves determines the topology of spacetime." *Journal of Mathematical Physics* **18**, 1399–1404. https://doi.org/10.1063/1.523436

[37] Lamport, L. (1978). "Time, clocks, and the ordering of events in a distributed system." *Communications of the ACM* **21**(7), 558–565. https://doi.org/10.1145/359545.359563

[38] Lamport, L., Shostak, R., and Pease, M. (1982). "The Byzantine generals problem." *ACM Transactions on Programming Languages and Systems* **4**(3), 382–401. https://doi.org/10.1145/357172.357176

[39] Fischer, M. J., Lynch, N. A., and Paterson, M. S. (1985). "Impossibility of distributed consensus with one faulty process." *Journal of the ACM* **32**(2), 374–382. https://doi.org/10.1145/3149.214121

[40] Dijkstra, E. W. (1974). "Self-stabilizing systems in spite of distributed control." *Communications of the ACM* **17**(11), 643–644. https://doi.org/10.1145/361179.361202

[41] Newman, M. H. A. (1942). "On theories with a combinatorial definition of equivalence." *Annals of Mathematics* **43**(2), 223–243. https://doi.org/10.2307/1968867

[42] Rovelli, C. (1996). "Relational quantum mechanics." *International Journal of Theoretical Physics* **35**, 1637–1678. https://doi.org/10.1007/BF02302261

[43] Connes, A., and Rovelli, C. (1994). "Von Neumann algebra automorphisms and time-thermodynamics relation in generally covariant quantum theories." *Classical and Quantum Gravity* **11**, 2899–2917. https://doi.org/10.1088/0264-9381/11/12/007

[44] Zurek, W. H. (2003). "Decoherence, einselection, and the quantum origins of the classical." *Reviews of Modern Physics* **75**, 715–775. https://doi.org/10.1103/RevModPhys.75.715

[45] 't Hooft, G. (2016). *The Cellular Automaton Interpretation of Quantum Mechanics*. Fundamental Theories of Physics 185. Springer, Cham. https://doi.org/10.1007/978-3-319-41285-6

[46] Frauchiger, D., and Renner, R. (2018). "Quantum theory cannot consistently describe the use of itself." *Nature Communications* **9**, 3711. https://doi.org/10.1038/s41467-018-05739-8

[47] Ratcliffe, J. G. (2019). *Foundations of Hyperbolic Manifolds*, 3rd ed. Graduate Texts in Mathematics 149. Springer, Cham. https://doi.org/10.1007/978-3-030-31597-9

[48] Serre, J.-P. (1977). *Linear Representations of Finite Groups*. Graduate Texts in Mathematics 42. Springer, New York. https://doi.org/10.1007/978-1-4684-9458-7

[49] Cohan, N. V. (1958). "The spherical harmonics with the symmetry of the icosahedral group." *Proceedings of the Cambridge Philosophical Society* **54**(1), 28–38. https://doi.org/10.1017/S0305004100033156

[50] Bisognano, J. J., and Wichmann, E. H. (1975). "On the duality condition for a Hermitian scalar field." *Journal of Mathematical Physics* **16**, 985–1007. https://doi.org/10.1063/1.522605

[51] Bisognano, J. J., and Wichmann, E. H. (1976). "On the duality condition for quantum fields." *Journal of Mathematical Physics* **17**, 303–321. https://doi.org/10.1063/1.522898

[52] Wiesbrock, H.-W. (1993). "Half-sided modular inclusions of von Neumann algebras." *Communications in Mathematical Physics* **157**, 83–92. https://doi.org/10.1007/BF02098019

[53] Jacobson, T. (1995). "Thermodynamics of spacetime: The Einstein equation of state." *Physical Review Letters* **75**, 1260–1263. https://doi.org/10.1103/PhysRevLett.75.1260

[54] Jacobson, T. (2016). "Entanglement equilibrium and the Einstein equation." *Physical Review Letters* **116**, 201101. https://doi.org/10.1103/PhysRevLett.116.201101

[55] van der Bij, J. J., van Dam, H., and Ng, Y. J. (1982). "The exchange of massless spin-two particles." *Physica A* **116**, 307–320. https://doi.org/10.1016/0378-4371(82)90247-3

[56] Ellis, G. F. R., van Elst, H., Murugan, J., and Uzan, J.-P. (2011). "On the trace-free Einstein equations as a viable alternative to general relativity." *Classical and Quantum Gravity* **28**, 225007. https://doi.org/10.1088/0264-9381/28/22/225007

[57] Van Raamsdonk, M. (2010). "Building up spacetime with quantum entanglement." *General Relativity and Gravitation* **42**, 2323–2329. https://doi.org/10.1007/s10714-010-1034-0

[58] Hall, B. C. (2015). *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, 2nd ed. Graduate Texts in Mathematics 222. Springer, Cham. https://doi.org/10.1007/978-3-319-13467-3

[59] Doplicher, S., and Roberts, J. E. (1989). "A new duality theory for compact groups." *Inventiones Mathematicae* **98**, 157–218. https://doi.org/10.1007/BF01388849

[60] Doplicher, S., and Roberts, J. E. (1990). "Why there is a field algebra with a compact gauge group describing the superselection structure in particle physics." *Communications in Mathematical Physics* **131**, 51–107. https://doi.org/10.1007/BF02097680

[61] Georgi, H., and Glashow, S. L. (1974). "Unity of all elementary-particle forces." *Physical Review Letters* **32**(8), 438–441. https://doi.org/10.1103/PhysRevLett.32.438

[62] Bouchiat, C., Iliopoulos, J., and Meyer, Ph. (1972). "An anomaly-free version of Weinberg's model." *Physics Letters B* **38**(7), 519–523. https://doi.org/10.1016/0370-2693(72)90532-1

[63] Tong, D. (2017). "Line operators in the Standard Model." *Journal of High Energy Physics* **07**, 104. https://doi.org/10.1007/JHEP07(2017)104

[64] Witten, E. (1982). "An SU(2) anomaly." *Physics Letters B* **117**, 324–328. https://doi.org/10.1016/0370-2693(82)90728-6

[65] Mohr, P. J., Newell, D. B., Taylor, B. N., and Tiesinga, E. (2025). "CODATA recommended values of the fundamental physical constants: 2022." *Reviews of Modern Physics* **97**, 025002. https://doi.org/10.1103/RevModPhys.97.025002

[66] Mueller, B., Poneder, M., Kim, J., and Friedman, D. A. (2026). *An Executable Observer for Observer Patch Holography: Specification, Executed Components and Ordered Joins*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/executable_observer_construction.pdf

[67] FloatingPragma (2026). The runnable OPH model: observer implementation, tests, receipts and construction guide. https://github.com/FloatingPragma/observer-patch-holography/tree/r2055/model

[68] FloatingPragma (2026). Admissible record laws for native repair. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/record_laws/README.md

[69] FloatingPragma (2026). Native quantum mixing and the full-carrier memory boundary. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/QUANTUM_MIXING.md

[70] FloatingPragma (2026). What complete native records can recover. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/RECORD_RECOVERY.md

[71] FloatingPragma (2026). The infinite-occupation sharp-readout history of the common interacting source. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/INFINITE_HISTORY.md

[72] FloatingPragma (2026). Complete comparisons with the actual finite pointer. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/READOUT.md

[73] FloatingPragma (2026). The finite executable observer specification. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/COMPOSITION.md

[74] FloatingPragma (2026). Coherent native repairs around one continuing quantum history. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/QUANTUM_NATIVE.md

[75] FloatingPragma (2026). Active encoded memory in the continuing observer. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/ENCODED_MEMORY.md

[76] FloatingPragma (2026). One process, overlapping observer records and an effective transfer. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer_correspondence/README.md

[77] FloatingPragma (2026). Source-compiled apparatus in a complete observer-clock history. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/compiled_clock_history/README.md

[78] FloatingPragma (2026). Joint local clocks and quantum fields with explicit memory. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/joint_clock_process/README.md

[79] FloatingPragma (2026). Fixed observer predictions and complete-count comparison: modular clock recovery and native repair response. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/predictions/README.md

[80] FloatingPragma (2026). Identifying the complete current-resolving repair law. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/repair_identification/README.md

[81] FloatingPragma (2026). A local action for native quantum observations. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/native_action/README.md

[82] FloatingPragma (2026). Coherent observer outputs on matched source screens. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/coherent_screen_observer/README.md

[83] FloatingPragma (2026). A spatial fluctuation field from one native history: the common spatial generator, its locality and spectral dimension, and the boundary-trace test. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/physical_space_time/OCCUPATION_FIELD.md

[84] FloatingPragma (2026). Joint time and field laws on matched source screens. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer_time_fields/REFINEMENT.md

[85] FloatingPragma (2026). Tangent first laws and finite entropy changes: native reference entropy and region boundaries. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/code/geometry/ENTROPY_FIRST_LAW_AUDIT.md

[86] FloatingPragma (2026). Exact native causal cones. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/CAUSALITY.md

[87] FloatingPragma (2026). A finite sequential native-law prediction protocol. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/closed_prediction_protocol/README.md

[88] FloatingPragma (2026). Classical record responses and published Bell measurements. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/predictions/BELL_RECORDS.md

[89] FloatingPragma (2026). OPH predictions and public-data comparisons. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/docs/research/PUBLIC_DATA_PREDICTION_AUDIT.md

[90] Hensen, B. et al. (2015). Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres. *Nature* 526, 682–686. https://doi.org/10.1038/nature15759

[91] King, S. A. et al. (2022). An optical atomic clock based on a highly charged ion. *Nature* 611, 43–47. https://doi.org/10.1038/s41586-022-05245-4

[92] Louis, T. et al. (2025). The Atacama Cosmology Telescope: DR6 power spectra, likelihoods and ΛCDM parameters. arXiv:2503.14452. https://arxiv.org/abs/2503.14452

[93] Mosco, U. (1994). Composite media and asymptotic Dirichlet forms. *J. Funct. Anal.* 123, 368–421.

[94] Fukushima, M., Oshima, Y. and Takeda, M. (2011). *Dirichlet Forms and Symmetric Markov Processes*, 2nd edition. De Gruyter, Berlin.

[95] Li, X. and Ding, P. (2017). General forms of finite population central limit theorems with applications to causal inference. *J. Amer. Statist. Assoc.* 112, 1759–1769.

[96] Coulhon, T. and Sikora, A. (2008). Gaussian heat kernel upper bounds via the Phragmén–Lindelöf theorem. *Proc. London Math. Soc.* 96, 507–544.

[97] Girouard, A. and Polterovich, I. (2017). Spectral geometry of the Steklov problem. *J. Spectral Theory* 7, 321–359.

[98] FloatingPragma (2026). *Selecting one source, interface and repair law.* Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/SOURCE_SELECTION.md

[99] FloatingPragma (2026). *The OPH Basis*: the closure hypothesis, the no-outside rule, the agreement condition, the executable specification, and the theorem-library premises. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/docs/BASIS.md

[100] Wheeler, J. A. (1983). "Law without law." In J. A. Wheeler and W. H. Zurek (eds.), *Quantum Theory and Measurement*, 182–213. Princeton University Press.

[101] Tegmark, M. (2008). "The mathematical universe." *Foundations of Physics* **38**, 101–150. https://doi.org/10.1007/s10701-007-9186-9

[102] Nozick, R. (1981). *Philosophical Explanations.* Harvard University Press.

[103] Lawvere, F. W. (1969). "Diagonal arguments and cartesian closed categories." *Lecture Notes in Mathematics* **92**, 134–145. https://doi.org/10.1007/BFb0080769

[104] A. D. Wissner-Gross (2026). *A Parameter-Free Candidate for the Fine-Structure Constant from Entangled Qudits.* Author preprint, dated October 23, 2026. https://alexwg.org/publications/ChainTheoryAlpha.pdf

[105] Colladay, D., and Kostelecký, V. A. (1998). "Lorentz-violating extension of the standard model." *Physical Review D* **58**, 116002. https://doi.org/10.1103/PhysRevD.58.116002

[106] Conway, J. H., and Sloane, N. J. A. (1999). *Sphere Packings, Lattices and Groups.* Springer New York. https://doi.org/10.1007/978-1-4757-6568-7

[107] Delsarte, P., Goethals, J. M., and Seidel, J. J. (1977). "Spherical codes and designs." *Geometriae Dedicata* **6**, 363–388. https://doi.org/10.1007/BF03187604

[108] Godsil, C., and Royle, G. (2001). *Algebraic Graph Theory.* Springer. https://doi.org/10.1007/978-1-4613-0163-9

[109] Jacobson, T., Liberati, S., and Mattingly, D. (2003). "Threshold effects and Planck scale Lorentz violation: Combined constraints from high energy astrophysics." *Physical Review D* **67**, 124011. https://doi.org/10.1103/PhysRevD.67.124011

[110] Kostelecký, V. A., and Mewes, M. (2009). "Electrodynamics with Lorentz-violating operators of arbitrary dimension." *Physical Review D* **80**, 015020. https://doi.org/10.1103/PhysRevD.80.015020

[111] Kostelecký, V. A., and Russell, N. (2011). "Data tables for Lorentz and CPT violation." *Reviews of Modern Physics* **83**, 11–31. https://doi.org/10.1103/RevModPhys.83.11

[112] Meyer, B. (1954). "On the Symmetries of Spherical Harmonics." *Canadian Journal of Mathematics* **6**, 135–157. https://doi.org/10.4153/CJM-1954-016-2

[113] Strikwerda, J. C. (2004). *Finite Difference Schemes and Partial Differential Equations.* SIAM. https://doi.org/10.1137/1.9780898717938

[114] Trefethen, L. N. (1982). "Group Velocity in Finite Difference Schemes." *SIAM Review* **24**, 113–136. https://doi.org/10.1137/1024038

[115] Weinberg, S. (1995). *The Quantum Theory of Fields, Volume 1: Foundations.* Cambridge University Press. https://doi.org/10.1017/CBO9781139644167

AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.